

Urinalysis and Body Fluids

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7.1

Routine Physical and Biochemical Urine Tests

1. Which statement regarding renal function is true?

- A. Glomeruli are far more permeable to H₂O and salt compared with other capillaries
- B. The collecting tubule reabsorbs sodium and secretes potassium in response to antidiuretic hormone (ADH)
- C. The collecting tubule is permeable to H₂O only in the presence of aldosterone
- D. The thick ascending limb is highly permeable to and urea

Body fluids/Apply knowledge of fundamental biological characteristics/Urine/1

2. Which statement regarding normal salt and H₂O handling by the nephron is correct?

- A. The thick ascending limb of the tubule is highly permeable to salt but not H₂O
- B. The stimulus for ADH release is low arterial pressure in the afferent arteriole
- C. The descending limb of the tubule is impermeable to urea but highly permeable to salt
- D. Renin is released in response to high plasma osmolality

Body fluids/Apply knowledge of fundamental biological characteristics/Urine/1

3. Which statement concerning renal tubular function is true?

- A. In salt deprivation, the kidneys will conserve sodium at the expense of potassium
- B. Potassium is not excreted when serum concentration is less than 3.5 mmol/L
- C. No substance can be excreted into urine at a rate that exceeds the glomerular filtration rate (GFR)
- D. When tubular function is lost, the specific gravity (SG) of urine will be below 1.005

Body fluids/Correlate laboratory data with physiological processes/Urine electrolytes/2

Answers to Questions 1–3

- 1. **A** The formation of plasma ultrafiltrate depends on high hydrostatic pressure and permeability of the glomeruli. Aldosterone is released when afferent arterial pressure falls, and ADH is released when plasma osmolality becomes too high. The collecting tubule reabsorbs sodium and secretes potassium in response to aldosterone and is permeable to H₂O only in the presence of ADH. The thick ascending limb is permeable to salt, but not to H₂O or urea.
- 2. **A** The tubules are able to concentrate the filtrate because the descending limb is highly permeable to H₂O and urea, but not to salt, and the ascending limb is permeable to salt. Salt leaving the thick ascending limb creates a hypertonic interstitium that forces H₂O from the descending limb. Renin is released in response to low hydrostatic pressure in the afferent arteriole, which stimulates the juxtaglomerular cells. ADH is released by the posterior pituitary in response to high plasma osmolality.
- 3. **A** Sodium is a threshold substance; that is, no sodium will be excreted in urine until the

renal threshold (a plasma sodium concentration of approximately 120 mmol/L) is exceeded. Potassium is not a threshold substance and will be secreted by the tubules even when plasma potassium levels are low. Patients on diuretics or who have hypovolemia become hypokalemic for this reason. Some substances (e.g., penicillin) can be excreted at a rate exceeding glomerular filtration because the tubules secrete them. The tubules are responsible for concentrating the filtrate in conditions of water deprivation and diluting it in conditions of water excess. When tubular function is lost, salt and water equilibrate by passive diffusion, and the SG of urine becomes the same as that of plasma, approximately 1.010.

4. Which of the following is *inappropriate* when collecting urine for routine bacteriologic culture?

- A. The container must be sterile
- B. The midstream void technique must be used
- C. The collected sample must be plated within 2 hours unless refrigerated
- D. The sample may be held at 2°C to 8°C for up to 48 hours prior to plating

Body fluids/Apply knowledge to identify sources of error/Specimen collecting and handling/2

5. Which statement about sample collection for routine urinalysis is true?

- A. Preservative tablets should be used for collecting random urine specimens
- B. Containers may be washed and reused if rinsed in deionized H₂O
- C. Samples may be stored at room temperature for up to 2 hours
- D. A midday sample is preferred when renal disease is suspected

Body fluids/Apply knowledge to identify sources of error/Specimen collection and handling/2

6. Which urine color is correlated correctly with the pigment-producing substance?

- A. Smoky red urine with homogentisic acid
- B. Dark amber urine with myoglobin
- C. Deep yellow urine and yellow foam with bilirubin
- D. Red-brown urine with biliverdin

Body fluids/Correlate laboratory data with physiological processes/Urine color/2

7. Which of the following substances will cause urine to produce red fluorescence when examined under an ultraviolet lamp (360 nm)?

- A. Myoglobin
- B. Porphobilinogen (PBG)
- C. Urobilin
- D. Coproporphyrin

Body fluids/Correlate clinical and laboratory data/Urine porphyrins/2

8. Which of the following conditions is associated with normal urine color but produces red fluorescence when urine is examined with an ultraviolet (Wood) lamp?

- A. Acute intermittent porphyria
- B. Lead poisoning
- C. Erythropoietic porphyria
- D. Porphyria cutanea tarda

Body fluids/Correlate clinical and laboratory data/Urine color/2

4. **D** Urine specimens should be plated and incubated within 2 hours of collection (some laboratories use a 1-hour time limit) and within 24 hours if the sample is refrigerated at 2°C to 8°C immediately after collection. No additives are permitted when urine is collected for culture.
5. **C** The first morning voided sample is the most sensitive for screening purposes because formed elements are concentrated, but random samples are satisfactory because glomerular bleeding, albuminuria, and cast formation may occur at any time. Preservative tablets should be avoided because they may cause chemical interference with some dry reagent strip and turbidimetric protein tests. Changes in glucose, bilirubin, and urobilinogen can occur within 30 minutes of collection. Therefore, samples should be refrigerated if not tested within 2 hours.
6. **C** Homogentisic acid causes dark brown or black-colored urine. Myoglobin causes a red to red-brown color in urine, and biliverdin causes a green or yellow-green color. In addition to metabolic diseases and renal disease, abnormal color can be caused by drugs, dyes excreted by the kidneys, and natural or artificial food coloring.
7. **D** Myoglobin causes a positive test for blood but does not cause urine to fluoresce. PBG causes urine to become dark (orange to orange-brown) on standing but does not fluoresce. Uroporphyrin and coproporphyrin produce red or orange-red fluorescence. Unlike hemoglobin, porphyrins lack peroxidase activity. Urobilin is an oxidation product of urobilinogen. It turns the urine orange to orange-brown but does not produce fluorescence.
8. **B** Lead poisoning blocks the synthesis of heme, causing accumulation of PBG and coproporphyrin III in urine. However, uroporphyrin levels are not sufficiently elevated to cause red pigmentation of urine. There is sufficient coproporphyrin to cause a positive test for fluorescence. Acute intermittent porphyria produces increased urinary delta-aminolevulinic acid (Δ -ALA), and PBG. The PBG turns urine orange to orange-brown upon standing. Erythropoietic porphyria and porphyria cutanea tarda produce large amounts of uroporphyrin, causing the urine to be red or port wine colored.

9. Which statement regarding porphyria is accurate?

- A. Porphyria is exclusively inherited
- B. All types cause an increase in urinary porphyrins
- C. All types are associated with anemia
- D. Serum, urine, and fecal tests may be needed for diagnosis

Body fluids/Apply principles of special procedures/Porphyrins/2

10. Which is the *most* common form of porphyria?

- A. Erythropoietic porphyria
- B. Acute intermittent porphyria
- C. Variegate porphyria
- D. Porphyria cutanea tarda

Body fluids/Apply knowledge disease states/Porphyria/2

11. Which of the following methods is the *least* sensitive and specific for measuring PBG in urine?

- A. Watson-Schwartz test

- B. LC-MS
- C. Ion exchange chromatography–Ehrlich reaction
- D. Isotope dilution–MS

Body fluids/Apply principles of special procedures/Urine PBG/2

12. A brown or black pigment in urine can be caused by:

- A. Gantrisin (pyridium)
- B. Phenolsulfonphthalein (PSP)
- C. Rifampin
- D. Melanin

Body fluids/Correlate clinical and laboratory data/Urine color/2

13. Urine that is dark red or port wine-colored may be caused by:

- A. Lead poisoning
- B. Porphyria cutanea tarda
- C. Alkaptonuria
- D. Hemolytic anemia

Body fluids/Correlate clinical and laboratory data/Urine color/2

Answers to Questions 9–13

- 9. D** Porphyria may be inherited as a result of an enzyme defect in heme synthesis or may be acquired as a result of lead poisoning, liver failure, or drug toxicity. The inherited porphyrias consist of eight subgroups based on which enzyme is deficient. They are divided clinically into three groups: neuropsychiatric, cutaneous, or mixed. The neurological porphyrias are not associated with anemia, but erythropoietic porphyria, a type of cutaneous porphyria, is. In general, neurological porphyrias are associated with increases in PBG and Δ -aminolevulinic acid (porphyrin precursors), whereas cutaneous porphyrias are associated with increased urinary porphyrins. No one sample type can be used to identify all subgroups, and sometimes all three are needed.
- 10. D** Porphyria is a rare condition, although most of the inherited forms are autosomal dominant. Porphyria cutanea tarda results from a deficiency of uroporphyrinogen decarboxylase, and hence, the carboxylated forms of uroporphyrin accumulate in plasma and spill into urine. The enzyme in hepatocytes is susceptible to drugs, alcohol, and hepatitis, which trigger the disease. The disease usually appears in middle-aged adults, the majority of whom have hepatitis C infection. The uroporphyrins are highly fluorescent and may cause port wine-colored urine. Affected persons present with skin blisters and skin burns if they have been exposed to sunlight.
- 11. A** The Watson-Schwartz test is a qualitative screening test for PBG and is based on the principle that dietary indole compounds and urobilinogen can be separated from PBG by extraction. PBG is extracted in n-butanol, whereas urobilinogen and dietary indoles are extracted into chloroform. However, the sensitivity and specificity of the test are poor in comparison with chromatographic and mass spectroscopic methods that better separate PBG from interfering substances. PBG is elevated in neurological porphyrias, the most common of which is acute intermittent porphyria.
- 12. D** Excretion of melanin in malignant melanoma and homogentisic acid in alkaptonuria cause urine to turn black on standing. Other substances that may cause brown or black-

colored urine are methemoglobin, PBG, porphobilin, and urobilin. Gantrisin, PSP dye, and rifampin are three examples of drugs that cause red or orange-red urine.

- 13. B** Porphyrria cutanea tarda and erythropoietic porphyria produce sufficient uroporphyrins to cause dark red urine. Acute intermittent porphyria produces large amounts of PBG, which may be oxidized to porphobilin, turning the urine orange to orange-brown.

- 14. Which of the following tests is affected *least* by standing or improperly stored urine?**

A. Glucose
B. Protein
C. pH
D. Bilirubin

Body fluids/Apply knowledge to identify sources of error/Urine/Specimen collection and handling/2

- 15. Which one of the following characteristics would be a reason for performing a microscopic examination of urinary sediment?**

A. High volume
B. Color intensity
C. Turbidity
D. Specimen from a Foley catheter

Body fluids/Apply principles of basic laboratory procedures/Urine/Specimen collection and handling/2

- 16. Which of the following is *appropriate* when collecting a 24-hour urine sample for metanephrines?**

A. Urine in the bladder is voided at the start of the test and added to the collection container
B. At 24 hours, any urine in the bladder is voided and discarded
C. All urine should be collected in a single container that is kept refrigerated
D. Ten milliliters of 1N sodium hydroxide should be added to the container before collection

Body fluids/Apply knowledge to identify sources of error/Urinary catecholamines/2

- 17. Urine production of less than 400 mL/day is:**

A. Consistent with normal renal function and H₂O balance
B. Termed isosthenuria
C. Defined as oliguria
D. Associated with diabetes mellitus

Body fluids/Correlate clinical and laboratory data/Urine volume/2

- 18. Which of the following contributes to SG, but not to osmolality?**

A. Protein
B. Salt
C. Urea
D. Glucose

Body fluids/Evaluate laboratory data to determine possible inconsistent results/Specific gravity/2

14. **B** Standing urine may become alkaline because of loss of volatile acids and ammonia production. Bilirubin glucuronides may become hydrolyzed to unconjugated bilirubin or oxidized to biliverdin, resulting in a false-negative result on the dry reagent strip test. Glucose can be consumed by glycolysis or oxidation by cells.
15. **C** Urine microscopy reflex testing is used by laboratories that do not perform automated microscopic urinalysis because of the low likelihood of significant findings when the microscopic analysis is performed on samples with normal physical characteristics and biochemical results. Urine samples with abnormal color and clarity are usually included in the criteria for reflex testing because either may be associated with abnormal cells or crystals.
16. **C** When collecting a 24-hour urine sample, the bladder must be emptied of urine at the start of the test and discarded. The bladder must be emptied at the conclusion of the test and the urine added to the collection. To prevent degradation of the catecholamines vanillylmandelic acid (VMA), metanephrines, and cortisol, urine must be refrigerated during storage and kept at 2°C to 8°C until analysis or frozen. If the pH of the specimen exceeds 3.0, degradation of catecholamines can occur. Preservation of urine with acid is no longer required for VMA, metanephrines, and cortisol.
17. **C** Normal daily urine excretion is usually 600 to 1,600 mL/day. *Isosthenuria* refers to urine of constant SG of 1.010, which is the SG of the glomerular filtrate. Glycosuria causes retention of H₂O within the tubule, resulting in dehydration and polyuria, rather than oliguria.
18. **A** All substances that dissolve in the urine contribute to osmotic pressure or osmolality. This includes nonionized solutes such as urea, uric acid, and glucose as well as salts, but not colloids, such as protein and lipids.

19. Urine with an SG consistently between 1.002 and 1.003 indicates:

- A. Acute glomerulonephritis
- B. Renal tubular failure
- C. Diabetes insipidus
- D. Addison disease

Body fluids/Evaluate laboratory data to recognize health and disease states/Specific gravity/2

20. In which of the following conditions is the urine SG likely to be below 1.025?

- A. Diabetes mellitus
- B. Drug overdose
- C. Chronic renal failure
- D. Prerenal failure

Body fluids/Evaluate data to recognize health and disease states/Specific gravity/2

21. Which statement regarding methods for measuring SG is true?

- A. Refractometry is the most accurate way to determine dissolved solute concentration
- B. Colorimetric SG test results are falsely elevated when a large quantity of glucose is present
- C. Colorimetric SG readings are falsely low when pH is alkaline
- D. Refractometry should be performed before the urine is centrifuged

Body fluids/Apply knowledge to identify sources of error/Specific gravity/2

22. What is the principle of the colorimetric reagent strip determination of SG in urine?

- A. Ionic strength alters the pK_a of a polyelectrolyte
- B. Sodium and other cations are chelated by a ligand that changes color
- C. Anions displace a pH indicator from a mordant, making it water soluble
- D. Ionized solutes catalyze oxidation of an azo dye

Body fluids/Apply principles of basic laboratory procedures/Specific gravity/1

23. Which statement regarding urine pH is true?

- A. A high-protein diet promotes an alkaline urine pH
- B. pH tends to decrease as urine is stored
- C. Contamination should be suspected if urine pH is less than 4.5
- D. Bacteriuria is most often associated with a low urine pH

Body fluids/Correlate clinical and laboratory data/Urine pH/2

Answers to Questions 19–23

- 19. C In severe renal diseases, the tubules fail to concentrate the filtrate. Salt and H_2O equilibrate by diffusion, causing an SG of about 1.010. If the SG of urine is below that of plasma, free H_2O is lost. This results from failure to produce ADH (inherited diabetes insipidus) or from failure of the tubules to respond to ADH (nephrogenic diabetes insipidus, which can be caused by drugs, polycystic kidney disease, and hypercalcemia).
- 20. C Glucose and drug metabolites increase the SG of urine. In prerenal failure, the tubules are undamaged. Ineffective arterial pressure stimulates aldosterone release. This increases sodium reabsorption, which stimulates ADH release. Water and salt are retained, and the urine:plasma osmolar (U:P) ratio exceeds 2:1. Chronic renal failure is associated with nocturia, polyuria, and low SG caused by scarring of the collecting tubules.
- 21. C Cells and undissolved solutes refract light and will cause a falsely high specific gravity reading by refractometry if urine is not centrifuged. Colorimetric SG tests are less sensitive to nonionized compounds, such as urea and glucose, and are negatively biased when large quantities of nonelectrolytes are present. Colorimetric SG readings are determined by a pH change on the test pad and are approximately 0.005 lower when pH is 6.5 or higher. Osmometry is the most accurate way to measure the concentration of dissolved solutes in urine.
- 22. A A polyelectrolyte with malic acid residues will ionize in proportion to the ionic strength of urine. This causes the pH indicator, bromthymol blue, to react as if it were in a more acidic solution. The indicator will be blue at low SG and green at higher SG.
- 23. C Bacteriuria is usually associated with an alkaline pH caused by the production of ammonia from urea. Extended storage may result in loss of volatile acids, causing increased pH. A high-protein diet promotes excretion of inorganic acids. The tubular maximum for hydrogen ion secretion occurs when urine pH reaches 4.5, the lowest urinary pH that the kidneys can produce.

24. In renal tubular acidosis, the pH of urine is:

- A. Consistently acid
- B. Consistently alkaline
- C. Neutral
- D. Variable, depending on diet

Body fluids/Correlate clinical and laboratory data/Urine pH/2

25. The normal daily urine output for an adult is approximately:

- A. 0.2 to 0.5 L
- B. 0.6 to 1.6 L
- C. 2.7 to 3.0 L
- D. 3.2 to 3.5 L

Body fluids/Apply knowledge of fundamental biological characteristics/Urine/1

26. The SG of the filtrate in the Bowman space is approximately:

- A. 1.000 to 1.002
- B. 1.004 to 1.006
- C. 1.008 to 1.010
- D. 1.012 to 1.014

Body fluids/Apply knowledge of fundamental biological characteristics/Urine/1

27. A patient with partially compensated respiratory alkalosis would have a urine pH of:

- A. 4.5 to 5.5
- B. 5.5 to 6.5
- C. 6.5 to 7.5
- D. 7.5 to 8.5

Body fluids/Correlate clinical and laboratory data/Urine pH/2

28. Which of the following is most likely to cause a false-positive result on the dry reagent strip test for urinary protein?

- A. Urine of high SG
- B. Highly buffered alkaline urine
- C. Bence-Jones protein
- D. Salicylates

Body fluids/Apply knowledge to identify sources of error/Urinary protein/2

29. When testing for urinary protein with sulfosalicylic acid (SSA), which condition may produce a false-positive result?

- A. Highly buffered alkaline urine
- B. The presence of x-ray contrast media
- C. Increased urinary SG
- D. The presence of red blood cells (RBCs)

Body fluids/Apply knowledge to identify sources of error/Urinary protein/2

Answers to Questions 24–29

- 24. B** Renal tubular acidosis results from a defect in the renal tubular reabsorption of bicarbonate. Hydrogen ions are not secreted when bicarbonate ions are not reabsorbed. Wasting of sodium bicarbonate (NaHCO_3) and potassium bicarbonate (KHCO_3) results in alkaline urine and hypokalemia in association with acidosis.

25. **B** Under conditions of normal fluid intake, the reference range for urine volume is 0.6 to 1.6 L per day. Urine output will vary widely with fluid intake. In cases of fluid deprivation, almost all filtrate will be reabsorbed, resulting in daily excretion as low as 500 mL. When fluid intake is excessive, up to 2.0 L of urine may be voided. Urine output beyond these extremes is considered abnormal.
26. **C** The SG of the filtrate in the Bowman space approximates the SG of plasma because sodium, chloride, glucose, urea, and other main solutes are completely filtered by the glomeruli. This corresponds to an osmolality of approximately 280 mOsm/kg.
27. **D** Urine pH is determined by diet, acid–base balance, water balance, and renal function. In partially compensated respiratory alkalosis, the kidneys reabsorb less bicarbonate, which results in lower net acid excretion. The loss of bicarbonate helps compensate for alkalosis and causes urine pH to be alkaline.
28. **B** In addition to highly buffered alkaline urine, a false-positive protein result on the dry reagent strip test may be caused by quaternary ammonium compounds, which increase urine pH. Because the dry reagent strip tests are insensitive to globulins, a false-negative result is likely in the case of Bence-Jones proteinuria. Positive interference by drugs is uncommon for dry reagent strip protein tests but is common for turbidimetric tests. High urinary SG will suppress the color reaction of the strip protein tests.
29. **B** Turbidimetric assays are used to test urine suspected of giving a false-positive result on the dry reagent strip test for albumin because the urine is highly alkaline (pH greater than or equal to 8.0) or contains pigmentation that interferes with reading the protein test pad. In addition, SSA tests are used when screening urine for an increased concentration of globulins because dry reagent strip tests are far less sensitive to globulins. SSA is less specific but more sensitive for albuminuria than conventional dry reagent strip tests. Iodinated dyes, penicillin, salicylate, and tolbutamide may result in false-positive results. Trace turbidity is difficult to determine when urine is cloudy as a result of bacteriuria, mucus, or crystals. Alkaline urine may titrate SSA, reducing its sensitivity. For these reasons, the SSA test is not used routinely.

30. A discrepancy between the urine SG determined by measuring refractive index and urine osmolality would be *most likely* to occur:

- A. After catheterization of the urinary tract
- B. In diabetes mellitus
- C. After intravenous pyelography (IVP)
- D. In uremia

Body fluids/Evaluate data to determine possible inconsistent results/Specific gravity/2

31. Which of the following is *likely* to result in a false-negative dry reagent strip test for proteinuria?

- A. Penicillin
- B. Aspirin
- C. Amorphous phosphates
- D. Bence-Jones protein

Body fluids/Apply knowledge to identify sources of error/Urinary protein/1

32. Daily loss of protein in urine normally does not exceed:

- A. 30 mg
- B. 50 mg
- C. 100 mg
- D. 150 mg

Body fluids/Apply knowledge of fundamental biological characteristics/Urinary protein/1

33. Which of the following is *least* likely to cause a false-positive result on turbidimetric protein tests?

- A. Tolbutamide
- B. X-ray contrast media
- C. Penicillin or sulfa antibiotics
- D. Ascorbic acid

Body fluids/Apply knowledge to identify sources of error/Urinary protein/2

34. Which statement *best* describes the clinical utility of tests for microalbuminuria?

- A. Testing may detect early renal involvement in diabetes mellitus
- B. Microalbuminuria refers to a specific subfraction of albumin found only in persons with diabetic nephropathy
- C. A positive test result indicates the presence of orthostatic albuminuria
- D. Testing should be part of the routine urinalysis

Body fluids/Correlate clinical and laboratory data/Urinary protein/2

Answers to Questions 30–34

- 30. C** The IVP dye contains iodine and is highly refractile. This increases the refractive index of urine, causing falsely high measurement of solute concentration. The refractive index is affected by the size and shape of solutes and undissolved solids, such as protein. Osmolality is the most specific measure of total solute concentration because it is affected only by the number of dissolved solutes.
- 31. D** Dry reagent strip tests using tetrabromophenol blue or tetrachlorophenol tetrabromosulfophthalein are poorly sensitive to globulins and may not detect immunoglobulin light chains. Turbidimetric methods, such as 3% SSA, will often detect Bence-Jones protein but may give a false-positive reaction with penicillin, tolbutamide, salicylates, and x-ray contrast dyes containing iodine. Amorphous phosphates may precipitate in refrigerated urine, making interpretation of turbidimetric tests difficult.
- 32. D** Small amounts of albumin and other low molecular weight proteins, such as amylase, β -microglobulins, and immunoglobulin fragments, are excreted in urine. Proteinuria does not normally exceed 30 mg/dL or 150 mg/day. The detection limit of the SSA test to albumin is approximately 1.5 to 2.0 mg/dL, and for dry reagent strip tests, it is approximately 15 mg/dL. Therefore, trace positives by either method may occur in the absence of renal disease.
- 33. D** Ascorbic acid may reduce diazo salts used in the bilirubin and nitrite tests, and react with hydrogen peroxide in peroxidase reactions. Therefore, persons taking megadoses of ascorbic acid (vitamin C) may show negative interference with tests for glucose, blood, bilirubin, and nitrite. Ascorbate does not cause either a false-negative or positive reaction for protein.

34. A The microalbumin test is an assay for measuring urinary albumin concentration that has an increased sensitivity (detection limit below 15 mg/dL) and is recommended for persons who are at risk for chronic renal disease, especially persons with diabetes mellitus. In diabetes, an early sign of renal involvement is an increased rate of albumin excretion in the range of 20 to 200 µg/mL or in excess of 30 mg albumin per gram of creatinine. Results in this range are significant in the at-risk population, even though the dry reagent strip test for protein may yield negative results. Dry reagent strip tests for microalbuminuria are too sensitive for use in routine urinalysis but are useful in screening persons with diabetes and hypertension for increased urinary albumin excretion.

35. Dry reagent strip tests for microalbuminuria that compare albumin to creatinine determine the creatinine concentration based on which principle?

- A. Formation of a Cu^{+2} –creatinine complex
- B. Enzymatic assay using sarcosine oxidase and peroxidase
- C. Reaction of creatinine with alkaline sodium picrate
- D. Change in pH as creatinine is converted to creatine

Body fluids/Apply principles of special laboratory procedures/Urine protein/2

36. Which of the following conditions is least likely to be detected by dry reagent strip tests for proteinuria?

- A. Orthostatic albuminuria
- B. Chronic renal failure
- C. Pyelonephritis
- D. Renal tubular proteinuria

Body fluids/Apply principles of basic laboratory procedures/Urine protein/2

37. The normal renal threshold for glucose is:

- A. 70 to 85 mg/dL
- B. 100 to 115 mg/dL
- C. 130 to 145 mg/dL
- D. 165 to 180 mg/dL

Body fluids/Apply knowledge of fundamental biological characteristics/Urine glucose/1

38. In which of the following conditions is glycosuria most likely?

- A. Addison disease
- B. Hypothyroidism
- C. Pregnancy
- D. Hypopituitarism

Body fluids/Correlate clinical and laboratory data/Urine glucose/2

39. In addition to ascorbate, the glucose oxidase reaction may be inhibited by which substance?

- A. Acetoacetic acid (AAA)
- B. ε-Aminocaproic acid
- C. Creatinine
- D. Azopyridium

Body fluids/Apply knowledge to identify sources of error/Urine glucose/1

Answers to Questions 35–39

- 35. A** The dry reagent strip test for creatinine contains anhydrous buffered copper II sulfate (CuIISO_4), alcoholic tetramethylbenzidine, and diisopropyl benzene dihydroperoxide. In the presence of creatinine, a copper–creatinine complex forms. This catalyzes the oxidation of a benzidine derivative by an alcoholic peroxide, forming a blue color on the test pad. Color intensity is proportional to creatinine concentration. Negative interference occurs from ascorbate and ethylenediaminetetraacetic acid (EDTA), which chelates the copper. Positive interference occurs from hemoglobin and some drugs (e.g., nitrofurantoin). The microalbumin concentration is determined by the protein error of indicator effect using a dye with increased sensitivity, bis-tetrabromosulfonephthalein.
- 36. D** The detection limit (sensitivity) of dry reagent strip protein tests is approximately 15 mg/dL albumin and is sufficient to detect urinary albumin levels found in orthostatic albuminuria and renal diseases, with the exception of tubular proteinuria. Renal tubular proteinuria results from failure of damaged tubules to reabsorb small proteins, such as β_2 -microglobulin. Dry reagent strip tests for proteinuria are poorly sensitive to globulins and do not detect small quantities of hemoglobin, myoglobin, or microglobulins. Electrophoresis of urine is used to detect renal tubular proteinuria.
- 37. D** The renal threshold is the concentration of a substance (e.g., glucose) in blood that must be exceeded before it can be detected in urine. Threshold substances require a carrier to transport them from the tubular lumen to the vasa recta. When the carrier becomes saturated, the tubular maximum is reached, causing the substance to be excreted in urine.
- 38. C** In addition to diabetes mellitus, glycosuria may occur in other endocrine diseases, in pregnancy, in response to drugs that affect glucose tolerance or renal threshold, and in several other conditions, especially those involving the liver or the central nervous system (CNS). Cushing disease and hyperthyroidism cause impaired glucose tolerance and hyperglycemia. Increased estrogens produced in pregnancy lower the renal threshold for glucose and may impair glucose tolerance. Hyperpituitarism causes hyperglycemia mediated by increased release of growth hormone.
- 39. A** AAA and salicylates may inhibit the glucose oxidase reaction by the same mechanism as ascorbate. These reducing agents compete with the chromogen for hydrogen peroxide. Low SG may increase and high SG decrease the color reaction for glucose in urine.

40. A positive glucose oxidase test and a negative test for reducing sugars indicates:

- A. True glycosuria
- B. False-positive reagent strip test result
- C. False-negative reducing test result caused by ascorbate
- D. Galactosuria

Body fluids/Evaluate laboratory data to determine possible inconsistent results/Urine glucose/2

41. A negative glucose oxidase test and a positive test for reducing sugars in urine indicates:

- A. True glycosuria
- B. A false-negative glucose oxidase reaction
- C. The presence of a nonglucose reducing sugar, such as galactose
- D. A trace quantity of glucose

Body fluids/Evaluate laboratory data to determine possible inconsistent results/Urine glucose/2

42. In what condition may urinary ketone tests underestimate ketosis?

- A. Acidosis
- B. Hemolytic anemia
- C. Renal failure
- D. Excessive use of vitamin C

Body fluids/Apply knowledge to identify sources of error/Urinary ketones/2

43. AAA is detected in urine by reaction with:

- A. Sodium nitroprusside
- B. o-Toluidine
- C. m-Dinitrobenzene
- D. m-Dinitrophenylhydrazine

Body fluids/Apply principles of basic laboratory procedures/Urinary ketones/1

44. Nondiabetic ketonuria can occur in all of the following except:

- A. Pregnancy
- B. Renal failure
- C. Starvation
- D. Lactate acidosis

Body fluids/Correlate clinical and laboratory data/Urinary ketones/2

Answers to Questions 40–44

- 40. A** Glucose oxidase is specific for β -D-glucose. Therefore, a positive reaction is always considered significant unless contamination is evident. A reducing test should not be used to confirm a positive result on the glucose oxidase test because this test is not as specific or as sensitive. Reducing sugar tests are used to screen infants for inborn errors of carbohydrate metabolism, such as galactosuria, but are not used to screen for glycosuria.
- 41. C** Reducing tests utilize alkaline copper sulfate and heat to oxidize glucose. Other reducing substances, including several sugars and antibiotics, may react, making the test inappropriate as a screening test for glucose. A positive test result for reducing sugars seen with a negative glucose oxidase test result may occur in lactose, galactose, and fructosuria and other disorders of carbohydrate metabolism. Such tests should always be confirmed with chromatography to identify the specific sugar and with assay of the specific enzyme involved.
- 42. A** Tests for urinary ketone bodies are sensitive to AAA. They react weakly with acetone and do not react with β -hydroxybutyric acid. Acidosis favors formation of β -hydroxybutyric acid and may cause a falsely low estimate of urine ketones in diabetic ketoacidosis. Ketonuria has many causes other than diabetic ketoacidosis, such as pregnancy, fever, protein calorie malnutrition, and dietary carbohydrate restriction.

Trace ketones tend to be more clinically significant when seen in urine with a low SG.

- 43. A** Urinary ketones are detected using alkaline sodium nitroprusside (nitroferricyanide). Mesna and other sulfhydryl compounds may cause a false-positive result on dry reagent strip tests and phenylpyruvic acid (PKU) and some antibiotics with the classic tube test.
- 44. B** Ketonuria results from excessive oxidation of fats forming acetyl coenzyme A (CoA). In addition to diabetes mellitus, ketonuria occurs in starvation, carbohydrate restriction, alkalosis, lactate acidosis, and von Gierke disease (glycogen stores cannot be utilized). Ketonuria also occurs in pregnancy, associated with increased vomiting and cyclic fever.

45. Which of the following statements regarding the classic nitroprusside reaction for ketones is true?

- A. The reaction is most sensitive to acetone
- B. Nitroprusside reacts with acetone, AAA, and β -hydroxybutyric acid
- C. It may be falsely positive in phenylketonuria
- D. The reaction is recommended for diagnosing ketoacidosis

Body fluids/Apply knowledge to identify sources of error/Urinary ketones/2

46. Hemoglobin in urine can be differentiated from myoglobin by using:

- A. 80% ammonium sulfate to precipitate hemoglobin
- B. Sodium dithionite to reduce hemoglobin
- C. o-Dianisidine instead of benzidine as the color indicator
- D. The dry reagent strip blood test

Body fluids/Select methods/Hemoglobinuria/2

47. Which of the following conditions is associated with a negative blood test result and an increase in urine urobilinogen?

- A. Calculi of the kidney or bladder
- B. Malignancy of the kidney or urinary system
- C. Crush injury
- D. Extravascular hemolytic anemia

Body fluids/Correlate clinical and laboratory data/Hematuria/2

48. Which statement about the dry reagent strip blood test is true?

- A. The test is based on the reaction of hemoglobin with peroxidase
- B. Abnormal color may be absent from the urine when the reaction is positive
- C. A nonhemolyzed trace is present when there are 1 to 2 RBCs per high-power field (RBCs/HPF)
- D. Salicylates cause a false-positive reaction

Body fluids/Apply principles of basic laboratory procedures/Hematuria/2

49. A moderately positive result on the blood test and trace protein test are seen on the dry reagent strip, and 11 to 20 RBCs/HPF are seen in the microscopic examination. These results are *most likely* caused by which of the following?

- A. Transfusion reaction
- B. Myoglobinuria
- C. Intravascular hemolytic anemia
- D. Recent urinary tract catheterization

Answers to Questions 45–49

45. **C** Tests for ketones are less sensitive to acetone than to AAA and do not detect β -hydroxybutyric acid. High levels of phenylpyruvic acid (phenylketonuria) will cause a false-positive reaction in the classic nitroprusside reaction but do not usually interfere with the dry reagent strip test for ketones. Serum ketones can be measured by gas chromatography, and β -hydroxybutyric acid can be measured enzymatically. The enzymatic assay for β -hydroxybutyrate in plasma is the recommended test for diagnosing ketoacidosis since acidosis favors its formation.
46. **A** Both hemoglobin and myoglobin have peroxidase activity and cause a positive blood test result. However, myoglobin is soluble in 80% weight per volume (w/v) ammonium sulfate in urine, but hemoglobin precipitates. A positive blood reaction with supernatant after addition of ammonium sulfate and sodium hydroxide (NaOH) confirms the presence of myoglobin. The presence of RBCs indicates that hemoglobin, rather than myoglobin, is present; however, the absence of RBCs does not rule out the presence of hemoglobin as the cause of a positive blood test result.
47. **D** A positive result on the blood test can occur from renal or lower urinary tract bleeding, intravascular hemolytic anemia, and transfusion reaction. Extravascular hemolysis results in increased bilirubin production, rather than in plasma hemoglobin. This may cause increased urobilinogen in urine, but not a positive blood reaction.
48. **B** The blood reaction uses anhydrous peroxide and tetramethylbenzidine. Hemoglobin has peroxidase activity and catalyzes the oxidation of tetramethylbenzidine by peroxide. The reaction is sensitive to submilligram levels of free hemoglobin, whereas visible hemolysis does not occur unless free hemoglobin exceeds 20 mg/dL. The test detects approximately 4–5 intact RBCs/HPF as a nonhemolyzed trace. Greater than 3 RBCs/HPF is abnormal.
49. **D** The blood test detects intact RBCs, hemoglobinuria, and myoglobinuria. Causes of hemoglobinuria include intravascular hemolytic anemias, transfusion reactions, and lysis of RBCs in the filtrate or urine caused by alkaline or hypotonic conditions. Causes of hematuria include acute and chronic glomerulonephritis, pyelonephritis, polycystic kidney disease, renal calculi, bladder and renal cancers, and catheterization of the urinary tract.

50. Which of the following results are discrepant?

- A. Small amount of blood, but negative protein
- B. Moderate amount of blood, but no RBCs in microscopic examination
- C. Negative blood, but 6 to 10 RBCs/HPF
- D. Negative blood, but positive protein

Body fluids/Apply knowledge to recognize sources of error/Hematuria/3

51. Which of the following statements regarding the dry reagent strip test for bilirubin is true?

- A. A positive test result is seen in prehepatic, hepatic, and posthepatic jaundice
- B. The test detects only conjugated bilirubin

- C. Standing urine may become falsely positive because of bacterial contamination
- D. High levels of ascorbate will cause positive interference

Body fluids/Apply knowledge to recognize sources of error/Urine bilirubin/2

52. Which of the following reagents is used to detect urobilinogen in urine?

- A. p-Dinitrobenzene
- B. p-Aminosalicylate
- C. p-Dimethylaminobenzaldehyde
- D. p-Dichloroaniline

Body fluids/Apply principles of basic laboratory procedures/Urine urobilinogen/1

Answers to Questions 50–52

- 50. C** The blood test detects as little as 0.015 mg/dL free hemoglobin and 4 to 5 RBCs/ μ L. The protein test detects 15 mg/dL albumin, but substantially more hemoglobin is required to obtain a positive test result. Therefore, a small blood reaction (nonhemolyzed or moderately hemolyzed trace, trace, or small) usually occurs in the absence of a positive protein. A positive blood test result often occurs in the absence of RBCs in the microscopic examination. This can result from intravascular hemolysis, myoglobinuria, or lysis of RBCs caused by alkaline or hypotonic urine. A positive test result for protein and a negative blood test result occur commonly in such conditions as orthostatic albuminuria, urinary tract infection, and diabetes mellitus. However, a negative blood test result should not occur if more than 3 to 4 RBCs/HPF are seen in the microscopic examination. Either the blood test is falsely negative (a missed nonhemolyzed trace), or yeast has been mistaken for RBCs.
- 51. B** Only the conjugated form of bilirubin is excreted into urine. Urinary bilirubin is positive in necrotic and obstructive jaundice but not in prehepatic jaundice, which results in a high level of serum unconjugated bilirubin. The highest levels of urinary bilirubin occur in obstructive jaundice, which causes decreased urinary urobilinogen. Very few drugs have been reported to interfere with urine bilirubin tests, which are based on the formation of azobilirubin by reaction with a diazonium salt. Positive interference by rifampin and chlorpromazine have been reported. Urine must be fresh because sunlight destroys bilirubin. Bacteria may cause hydrolysis of glucuronides, forming unconjugated bilirubin, which does not react with the diazonium reagent. Ascorbate inhibits the reaction by reducing the diazo reagent.
- 52. C** Urobilinogen reacts with the Ehrlich aldehyde reagent (p-dimethylaminobenzaldehyde in hydrochloric acid [HCl]) to form a pink color. Dry reagent strips use either p-dimethylaminobenzaldehyde or 4-methoxybenzene diazonium tetrafluoroborate to detect urobilinogen. The former reagent may react with PBG, salicylate, and sulfonamides giving falsely high results. False-positive results may occur in the presence of pyridium and gantrisin, which color urine orange-red. Formalin may cause a false-negative reaction.
- 53. Which of the following statements regarding urinary urobilinogen is true?**
- A. Diurnal variation occurs, and the highest levels are seen in the early morning
 - B. High levels accompanied by a positive bilirubin test result indicate obstructive jaundice

- C. Dry reagent strip tests do not detect decreased levels
- D. False-positive results may occur if urine is stored for more than 2 hours

Body fluids/Apply principles of basic laboratory procedures/Urine urobilinogen/2

54. Which of the following statements regarding the test for nitrite in urine is true?

- A. It detects more than 95% of clinically significant bacteriuria
- B. Formation of nitrite is unaffected by the urine pH
- C. The test is dependent on adequate dietary nitrate content
- D. A positive test differentiates bacteriuria from in vitro bacterial contamination

Body fluids/Apply knowledge to recognize sources of error/Nitrite/2

55. Which statement about the dry reagent strip test for leukocytes is true?

- A. The test detects only intact white blood cells (WBCs)
- B. The reaction is based on the hydrolysis of substrate by WBC esterases
- C. Several antibiotics may give a false-positive reaction
- D. The test is sensitive to 2 to 3 WBCs/HPF

Body fluids/Apply principles of basic laboratory procedures/Leukocytes/2

56. Which of the following statements about creatinine clearance is correct?

- A. Dietary restrictions are required during the 24 hours preceding the test
- B. Fluid intake must be restricted to below 600 mL in the 6 hours preceding the test
- C. Creatinine clearance is mainly determined by renal tubular function
- D. Creatinine clearance is dependent on lean body mass

Body fluids/Apply knowledge of fundamental biological characteristics/Creatinine clearance/1

Answers to Questions 53–56

- 53. C** Urobilinogen exhibits diurnal variation, and highest levels are seen in the afternoon. A 2-hour postprandial afternoon sample is the sample of choice for detecting increased urine urobilinogen. Urobilinogen is formed by bacterial reduction of conjugated bilirubin in the bowel. In obstructive jaundice, delivery of bilirubin into the intestine is blocked, resulting in decreased fecal, serum, and urine urobilinogen. However, the dry reagent strip tests are not sensitive enough to detect abnormally low levels. Urobilinogen is rapidly oxidized to urobilin, which does not react with dry reagent strip tests.
- 54. C** The nitrite test is dependent on the activity of bacterial reductase, and false-negative results have been reported when urine is highly acidic. Nitrite is formed by reduction of diet-derived nitrates and reacts with p-arsanilic acid or sulfanilamide to form a diazonium compound. This reacts with benzoquinoline to form a pink azo dye. False-negative results also occur in the presence of ascorbate, which reduces the diazonium product. Nitrite is positive in about 70% of clinically significant bacterial infections of the urinary tract. Sensitivity is limited by the requirements for dietary nitrate and 3- to 4-hour storage time in the bladder. In addition, the causative bacteria must be able to reduce nitrate.
- 55. B** Polymorphonuclear cells (PMNs) in urine are detected by the presence of esterases that hydrolyze an ester, such as indoxylcarbonic acid. The product reacts with a diazonium salt to give a purple color. The test detects esterases in urine as well as

intact WBCs but is not sensitive to less than 5 to 10 WBCs/HPF. Several antibiotics, high protein, and high SG inhibit the esterase reaction. Formalin may cause a false-positive result.

- 56. D** Although some creatinine is derived from the diet, it is rapidly filtered by the glomeruli, and time variations are reduced by collection of urine for 24 hours. Creatinine is produced from oxidation of creatine at a constant rate of about 2% per day. It is filtered completely and not significantly reabsorbed. However, creatinine secretion by the tubules is increased when filtrate flow is slow, and patients must be given at least 600 mL of H₂O at the start of the test and kept well hydrated throughout. Body size determines how much creatinine is produced, and clearance must be normalized to eliminate this variable. Because of the difficulty in collecting a valid 24-hour urine sample, the estimated glomerular filtration rate (eGFR) is recommended as the screening test for low GFR. Creatinine clearance should be used in situations where lean body mass may not be accurately represented by the patient's age and gender as used in the Modification of Diet in Renal Disease Study (MDRD) formula for eGFR.

- 57. A male patient's eGFR is 75 mL/min. This indicates:**

- A. Normal glomerular filtration rate
- B. The patient is uremic and will be hyperkalemic
- C. Renal tubular dysfunction
- D. Reduced glomerular filtration without uremia

Body fluids/Correlate clinical and laboratory data/Creatinine clearance/2

- 58. Which of the following substances can be used to calculate eGFR?**

- A. *p*-Aminohippuric acid (PAH)
- B. Glycine
- C. Cysteine
- D. Cystatin C

Body fluids/Correlate laboratory data with physiological processes/Renal function/1

- 59. Which statement regarding urea is true?**

- A. Urea is 100% filtered by the glomeruli
- B. Blood urea levels are independent of diet
- C. Urea is not significantly reabsorbed by the tubules
- D. Urea excretion is a specific measure of glomerular function

Body fluids/Correlate laboratory data with physiological processes/Urea/1

- 60. Given the following data, calculate the creatinine clearance.**

Serum creatinine = 1.2 mg/dL
Urine creatinine = 100 mg/dL
Urine volume = 1.4 L/day
Body surface area = 1.80 m²

- A. 47 mL/min
- B. 78 mL/min
- C. 100 mL/min
- D. 116 mL/min

Body fluids/Calculate/Creatinine clearance/2

Answers to Questions 57–60

- 57. D** Normal eGFR is 90 to 120 mL/min. Values below the lower reference limit, but greater than 60 mL/min, indicate glomerular damage but not of severity sufficient to cause symptoms or uremia.
- 58. D** Cystatin C is a small protease inhibitor that is produced at a constant rate, eliminated exclusively by glomerular filtration, and is not dependent on age, gender, or nutritional status. Plasma cystatin C is increased when the GFR is decreased, and levels can be used to give an eGFR in a similar manner to plasma creatinine. PAH is a substance that is completely filtered by the glomerulus and also secreted by the tubules. Amino acids are partially reabsorbed by the tubules and are not present in blood at a constant rate.
- 59. A** Blood urea nitrogen (BUN) is a sensitive indicator of renal disease but is not specific for glomerular function. BUN levels are affected by diet, hepatic function, tubular function, and filtrate flow, as well as the GFR. Although urea is completely filtered by the glomerulus, the tubules reabsorb 30% to 40% of the filtered urea, and this is why BUN concentration is higher than plasma creatinine. In prerenal failure, up to 70% of the filtered urea can be reabsorbed as a result of the slow movement of filtrate through the tubules. This causes BUN to rise much more than plasma creatinine in this condition. A BUN:creatinine ratio of 20:1 is highly suggestive of prerenal failure.
- 60. B** The clearance formula is $U \div P \times V \times 1.73/A$, where U = urine creatinine (mg/dL), P = plasma creatinine (mg/dL), V = urine volume (mL/min), and 1.73 = mean body surface area (m²):
- $$100 \text{ mg/dL} \div 1.2 \text{ mg/dL} \times 1.4 \text{ L/day} \times 1,000 \text{ mL/L} \times 1 \text{ day/1,440 min} \times (1.73 \text{ m}^2 \div 1.8 \text{ m}^2) = 78 \text{ mL/min}$$
- Note that the creatinine clearance is low (lower reference limit approximately 95 mL/min for males and 85 mL/min for females), although the serum creatinine is still within normal limits (0.5–1.2 mg/dL). The clearance test is more sensitive if done properly because as serum creatinine goes up, urine creatinine goes down.

Urine Microscopy and Clinical Correlations

1. Which of the following dyes are used to make Sternheimer-Malbin stain?

- A. Hematoxylin and eosin
- B. Crystal violet and safranin
- C. Methylene blue and eosin
- D. Methylene blue and safranin

Body fluids/Apply principles of basic laboratory procedures/Staining/1

2. Which of the following statements regarding WBCs in urinary sediment is true?

- A. “Glitter cells” seen in urinary sediment are a sign of renal disease
- B. Bacteriuria in the absence of WBCs indicates lower urinary tract infection (UTI)
- C. WBCs other than PMNs are not found in urinary sediment
- D. WBC casts indicate that pyuria is of renal, rather than lower urinary, origin

Body fluids/Correlate clinical and laboratory data/Urinary sediment/2

3. Which description of urinary sediment with the Sternheimer-Malbin stain is correct?

- A. Transitional epithelium: cytoplasm pale blue, nucleus dark blue
- B. Renal epithelium: cytoplasm light blue, nucleus dark purple
- C. Glitter cells: cytoplasm dark blue, nucleus dark purple
- D. Squamous epithelium: cytoplasm pink, nucleus pale blue

Body fluids/Apply knowledge of fundamental biological characteristics/Staining/2

Answers to Questions 1–3

- 1. B** The Sternheimer-Malbin stain is a supravital stain used to differentiate renal tubular epithelium from transitional cells and PMNs. The mononuclear cells are clearly distinguished from both live and dead PMNs. Transitional cells have pale blue cytoplasm, but renal cells take up both dyes, resulting in an azurophilic appearance (orange-purple cytoplasm and dark purple nucleus).
- 2. D** The majority of WBCs in urinary sediment will be PMNs. Eosinophils and mononuclear WBCs will occasionally be seen. High numbers of eosinophils often indicate an allergic drug reaction, causing inflammation in the medullary interstitium and tubules. Mononuclear cells are especially likely in patients with chronic inflammatory diseases and in renal transplant rejection, where they may account for as many as 30% of the WBCs. Glitter cells are PMNs with highly refractile granules exhibiting Brownian movement. They are seen only when the urine SG is below 1.020. These cells resist staining with the Sternheimer-Malbin stain and are considered to be living (fresh) WBCs. When seen in large numbers, they indicate urinary tract injury (with pseudopod extensions, they point to infection). The presence of bacteria in urine in the absence of PMNs usually results from contamination by vaginal or skin flora that multiply in vitro, especially in unrefrigerated specimens. The presence of WBC

casts is always significant and, when associated with pyuria and bacteriuria, indicates renal involvement in the infection.

- 3. A** After staining with the Sternheimer-Malbin stain, transitional epithelium are readily differentiated from renal tubular cells and WBCs because their cytoplasm is pale blue. Live WBCs exclude the Sternheimer-Malbin stain, whereas dead cells stain with a deeply blue-purple nucleus and pale orange-blue cytoplasm. Renal epithelium have an orange-purple cytoplasm and dark purple nucleus. Squamous epithelium have a blue or purple cytoplasm and an orange-purple nucleus. Red cells stain very pale pink or not at all and hyaline casts stain faintly pink.

- 4. SITUATION: A 5-mL urine specimen is submitted for routine urinalysis and is analyzed immediately. The SG of the sample is 1.012, and the pH is 6.5. The dry reagent strip blood test result is a large positive (3+), and the microscopic examination shows 11 to 20 RBCs/HPF. The leukocyte esterase reaction is a small positive (1+), and the microscopic examination shows 0 to 2 WBCs/HPF. What is the *most likely* cause of these results?**

- A. Myoglobin is present in the sample
- B. Free hemoglobin is present
- C. Insufficient volume is causing microscopic results to be underestimated
- D. Some WBCs have been misidentified as RBCs

Body fluids/Apply knowledge to identify sources of error/Urinalysis/3

- 5. Which of the following statements regarding epithelial cells in the urinary system is correct?**

- A. Caudate epithelial cells originate from the upper urethra
- B. Transitional cells originate from the upper urethra, ureters, bladder, or renal pelvis
- C. Cells from the proximal renal tubule are usually round
- D. Squamous epithelium line the vagina, urethra, and wall of the urinary bladder

Body fluids/Apply knowledge of fundamental biological characteristics/Urine sediment/2

- 6. Which of the statements regarding examination of unstained urinary sediment is true?**

- A. Renal cells can be differentiated reliably from WBCs
- B. Large numbers of transitional cells are often seen after catheterization
- C. Neoplastic cells from the bladder are not found in urinary sediment
- D. RBCs are easily differentiated from nonbudding yeast

Body fluids/Correlate clinical and laboratory data/Urine sediment/2

Answers to Questions 4–6

- 4. C** Given the SG and pH, most RBCs and WBCs will be intact. Both the RBC and WBC counts are lower than expected from the dry reagent strip test results. Myoglobin or free hemoglobin may account for the poor correlation between the blood reaction and the RBC count, but this does not explain the lower than expected WBC count. Microscopic reference ranges are based on concentrating a uniform volume of sediment from 12 mL of urine. When less urine is used, falsely low results will be obtained unless corrective action is taken. The specimen should be diluted with normal saline to 12 mL, then centrifuged at $450 \times g$ for 5 minutes. Urinary sediment should be prepared according to the established procedure and the results multiplied by the

dilution factor (in this case, $12 \div 5$, or 2.4).

5. **B** Caudate cells are transitional epithelium that have a sawtooth-shaped tail and are found in the urinary bladder and the pelvis of the kidney. Transitional epithelia line the upper two thirds of the urethra and the ureters as well as the urinary bladder and renal pelvis. Renal tubular cells may be columnar, polyhedral, or oval, depending on the portion of the tubule from which they originate. Cells from the proximal tubule are columnar and have a distinctive brush border. Squamous epithelia line the vagina and lower third of the urethra.
6. **B** Renal cells and PMNs are about the same size and can be confused in unstained sediment. Catheterization often releases large clumps or sheets of transitional and squamous cells. These should be distinguished from neoplastic cells derived from the urinary bladder. When cells appear atypical (e.g., large cells in metaphase), they should be referred to a pathologist for cytological examination. Nonbudding yeast cells are approximately the same in size and appearance as RBCs. When RBCs are reported in the absence of a positive blood test, the probability of an error in microscopy is high. The microscopic examination should be reviewed for the presence of yeast.

7. Which of the following statements regarding cells found in urinary sediment is true?

- A. Transitional cells resist swelling in hypotonic urine
- B. Renal tubular cells are often polyhedral and have an eccentric round nucleus
- C. Trichomonads have an oval shape with a prominent nucleus and a single anterior flagellum
- D. Clumps of bacteria are frequently mistaken for blood casts

Body fluids/Apply knowledge of fundamental biological characteristics/Urine sediment/2

8. Which of the following statements regarding RBCs in the urinary sediment is true?

- A. Yeast cells will lyse in dilute acetic acid but RBCs will not
- B. RBCs are often swollen in hypertonic urine
- C. RBCs of glomerular origin often appear dysmorphic
- D. Yeast cells will tumble when the cover glass is touched, but RBCs will not

Body fluids/Apply knowledge of fundamental biological characteristics/Urine sediment/2

9. Renal tubular epithelial cells are shed into urine in largest numbers in which condition?

- A. Malignant renal disease
- B. Acute glomerulonephritis
- C. Nephrotic syndrome
- D. Cytomegalovirus (CMV) infection of the kidney

Body fluids/Evaluate laboratory data to recognize health and disease states/Urinary sediment/2

10. The ova of which parasite is likely be found in the urinary sediment?

- A. *Trichomonas vaginalis*
- B. *Entamoeba histolytica*
- C. *Schistosoma hematobium*
- D. *Trichuris trichiura*

Body fluids/Apply knowledge of fundamental biological characteristics/Urinary sediment/1

11. Oval fat bodies are often seen in:

- A. Chronic glomerulonephritis

- B. Nephrotic syndrome
- C. Acute tubular nephrosis
- D. Renal failure of any cause

Body fluids/Correlate clinical and laboratory data/Urine sediment/2

Answers to Questions 7–11

7. **B** Transitional epithelial cells readily take up H₂O and appear much larger than renal cells or WBCs when urine is hypotonic. Transitional cells are considered a normal component of the sediment unless present in large numbers and associated with signs of inflammation, such as mucus and PMNs, or presenting features of malignant cells. In contrast, renal cells are significant when seen conclusively in the sediment. They are often teardrop-shaped, polyhedral, or elongated cells with a round eccentric nucleus. Conclusive identification requires staining. *T. vaginalis* displays an indistinct nucleus and two pairs of prominent anterior flagella. Amorphous urate crystals deposited on the slide may be mistaken for granular or blood casts.
8. **C** RBCs are difficult to distinguish from nonbudding yeast in unstained sediment. RBCs tumble when the cover glass is touched and will lyse when the sediment is reconstituted in normal saline containing 2% volume per volume (v/v) acetic acid. A nonhemolyzed trace blood reaction confirms the presence of RBCs. RBCs have a granular appearance in hypertonic urine as a result of crenation. The RBC membrane becomes distorted when passing through the glomerulus, often appearing scalloped, serrated, or invaginated. Such cells are called *dysmorphic RBCs* and are associated with glomerulonephritis.
9. **D** Although seen in glomerulonephritis and pyelonephritis, the largest numbers of renal tubular cells appear in urine in association with viral infections of the kidney. Renal epithelium may show characteristic viral inclusions associated with CMV and rubella. High numbers of renal epithelium are also found in the sediment of patients with drug-induced tubular nephrosis and in some cases of heavy metal poisoning. Renal tumors do not usually shed cells into urine.
10. **C** Ova of *S. hematobium* are most often recovered from urine because the adult trematodes colonize the blood vessels of the urinary bladder. The eggs are approximately 150 × 60 μm in size and are nonoperculated. They are yellowish and have a prominent terminal spine.
11. **B** Oval fat bodies are degenerated renal tubular epithelia that have reabsorbed cholesterol from the filtrate. Although they can occur in any inflammatory disease of the tubules, they are commonly seen in nephrotic syndrome, which is characterized by marked proteinuria and hyperlipidemia.
12. **Which statement regarding urinary casts is true?**
- A. Many hyaline casts may appear in urinary sediment after jogging or exercise
 - B. The finding of even a single cast indicates renal disease
 - C. Casts can be seen in significant numbers even when protein tests are negative
 - D. Hyaline casts will dissolve readily in acid urine

Body fluids/Apply knowledge to recognize sources of error/Urine casts/2

13. Which condition promotes the formation of casts in urine?

- A. Chronic production of alkaline urine
- B. Polyuria
- C. Reduced filtrate formation
- D. Low urine SG

Body fluids/Apply knowledge of fundamental characteristics/Urine casts/2

14. The mucoprotein that forms the matrix of a hyaline cast is called:

- A. Bence-Jones protein
- B. β -Microglobulin
- C. Tamm-Horsfall protein
- D. Arginine-rich glycoprotein

Body fluids/Apply knowledge of fundamental biological characteristics/Urine casts/1

15. “Pseudocasts” are often caused by:

- A. A dirty cover glass or slide
- B. Bacterial contamination
- C. Amorphous urates
- D. Mucus in urine

Body fluids/Apply knowledge to identify sources of error/Urine casts/2

16. Which of the following statements regarding urinary casts is correct?

- A. Fine granular casts are more significant than coarse granular casts
- B. Cylindruria is always clinically significant
- C. The appearance of cylindroids signals the onset of end-stage renal disease
- D. Broad casts are associated with severe renal tubular obstruction

Body fluids/Apply knowledge of fundamental biological characteristics/Urine casts/2

17. A sediment with moderate hematuria and RBC casts most likely results from:

- A. Chronic pyelonephritis
- B. Nephrotic syndrome
- C. Acute glomerulonephritis
- D. Lower urinary tract obstruction

Body fluids/Correlate clinical and laboratory data/Urine sediment/2

Answers to Questions 12–17

- 12. A** Proteinuria accompanies cylindruria because protein is the principal component of casts. After strenuous exercise, hyaline casts may be present in urinary sediment in significant numbers but will disappear after resting for at least 24 hours. Casts more readily dissolve in alkaline urine. The finding of a single cast, even a cellular cast, is not indicative of renal disease and should be considered a chance occurrence.
- 13. C** Cast formation is promoted by an acid filtrate, high solute concentration, slow movement of filtrate, and reduced filtrate formation. The appearance of a cast is dependent on the location and the time spent in the tubule, as well as the chemical and cellular composition of the filtrate.
- 14. C** Hyaline casts are composed of a mucoprotein called *Tamm-Horsfall protein*. In

addition, casts may contain cells, immunoglobulins, light chains, cellular proteins, fat, bacteria, and crystalloids.

15. **C** Pseudocasts are formed by amorphous urates that may deposit in uniform cylindrical shapes as the sediment settles under the cover glass. They may be mistaken for granular or blood casts. However, they are highly refractile and lack the well-defined borders of true casts.
16. **D** There is no clinical difference between fine and coarse granular casts. Granular casts may form by degeneration of cellular casts or from inclusion of urinary calculi, but some are of unknown etiology. *Cylindruria* refers to the presence of casts in the urine. Hyaline casts may be seen in small numbers in normal patients and in large numbers following strenuous exercise and long-distance running. Hyaline casts may also be increased in patients taking certain drugs, such as diuretics. Broad casts form in dilated or distal tubules and indicate severe tubular obstruction seen in chronic renal failure. Waxy casts form when there is prolonged stasis in the tubules and signal end-stage renal failure. Cylindroids are casts with tails and have no special clinical significance.
17. **C** RBC casts indicate the renal origin of hematuria. Urinary tract obstruction may be associated with hematuria from ruptured vessels, but not casts. WBCs and WBC casts predominate in pyelonephritis. Urinary sediment in chronic glomerulonephritis is variable but usually exhibits moderate to severe intermittent hematuria. In addition, pyuria and cylindruria (with granular, blood, broad, waxy, and epithelial casts) are frequent. In nephrotic syndrome, the sediment may be unremarkable except for the presence of oval fat bodies and hyaline casts. In some cases, fatty, waxy, and epithelial cell casts may also be found.

18. Urine sediment characterized by pyuria with bacterial and WBC casts indicates:

- A. Nephrotic syndrome
- B. Pyelonephritis
- C. Polycystic kidney disease
- D. Cystitis

Body fluids/Correlate clinical and laboratory data/Urine sediment/2

19. Which type of casts signals the presence of chronic renal failure?

- A. Blood casts
- B. Fine granular casts
- C. Waxy casts
- D. Fatty casts

Body fluids/Apply knowledge of fundamental biological characteristics/Urine casts/2

20. SITUATION: Urinalysis of a sample from a patient suspected of having a transfusion reaction reveals small, yellow-brown crystals in the microscopic examination. Dry reagent strip tests are normal with the exception of a positive blood reaction (moderate) and trace positive protein. The pH of urine is 6.5. What test should be performed to positively identify the crystals?

- A. Confirmatory test for bilirubin
- B. Cyanide–nitroprusside test
- C. Polarizing microscopy
- D. Prussian blue stain

Body fluids/Select course of action/Urine sediment/3

21. When examining urinary sediment, which of the following is considered an abnormal finding?

- A. 2 RBCs/HPF
- B. 1 hyaline cast per low-power field (LPF)
- C. 1 renal cell cast per LPF
- D. 5 WBCs/HPF

Body fluids/Evaluate laboratory data to recognize health and disease states/Urinary sediment/2

22. SITUATION: A urine sample with a pH of 6.0 produces an abundance of pink sediment after centrifugation that appears as densely packed yellow- to reddish-brown granules under the microscope. The crystals are so dense that no other formed elements can be evaluated. What is the *best* course of action?

- A. Request a new urine specimen
- B. Suspend the sediment in prewarmed saline, and then repeat centrifugation
- C. Acidify a 12-mL aliquot with three drops of glacial acetic acid, and heat to 56°C for 5 minutes before centrifuging
- D. Add five drops of 1N HCl to the sediment and examine

Body fluids/Select course of action/Urine sediment/3

Answers to Questions 18–22

- 18. B** Pyelonephritis results from bacterial infection of the renal pelvis and interstitium. It is characterized by polyuria resulting from failure of the tubules to reabsorb fluid. Compression and obstruction of the tubules by WBCs reduces H₂O reabsorption. The finding of WBC casts helps differentiate pyelonephritis from UTI.
- 19. C** Waxy casts form from the degeneration of cellular casts. Because the casts must remain lodged in the tubule long enough for the granular protein matrix to waxify, they are associated with chronic and end-stage renal failure. Both waxy and broad casts form in chronic renal failure when there is severe stasis, and they are associated with a poor prognosis.
- 20. D** A positive blood test result and trace protein occurring with a normal test for urobilinogen and an absence of RBCs are consistent with an intravascular transfusion reaction. Small, yellow-brown granular crystals at an acid pH may be uric acid, bilirubin, or hemosiderin. Bilirubin crystals are ruled out by the negative result on the dry reagent strip test for bilirubin. Potassium ferrocyanide is used in the Prussian blue staining reaction to detect hemosiderin deposits in urinary sediment. Hemosiderin is associated with hemochromatosis and increased RBC destruction. Causes of urinary hemosiderin include transfusion reaction, intravascular hemolytic anemia, and pernicious anemia.
- 21. C** Epithelial casts are rarely seen but indicate a disease process affecting the renal tubules. They are associated with diseases causing necrosis of the tubules, such as hepatitis, CMV and other viral infections, and mercury and ethylene glycol toxicity. Even occasional cellular casts are considered clinically significant.

22. B Urates are yellow-brown or reddish-brown granules and form in acid or neutral urine. They often form following refrigeration of urine and can be dissolved by addition of warm saline or dilute NaOH. Amorphous phosphates are colorless and form in neutral or alkaline urine. They dissolve in dilute acetic acid but precipitate if heated.

23. How can hexagonal uric acid crystals be distinguished from cystine crystals?

- A. Cystine is insoluble in hydrochloric acid, but uric acid is soluble
 - B. Cystine gives a positive nitroprusside test after reduction with sodium cyanide
 - C. Cystine crystals are more highly pigmented
 - D. Cystine crystals form at neutral or alkaline pH, uric acid forms at neutral to acidic pH
- Body fluids/Apply principles of special procedures/Urine crystals/2*

24. The presence of tyrosine and leucine crystals together in urinary sediment usually indicates:

- A. Renal failure
- B. Chronic liver disease
- C. Hemolytic anemia
- D. Hartnup disease

Body fluids/Correlate clinical and laboratory data/Urine crystals/2

25. Which of the following crystals is considered nonpathological?

- A. Hemosiderin
- B. Bilirubin
- C. Ammonium biurate
- D. Cholesterol

Body fluids/Evaluate laboratory data to recognize health and disease states/Urine crystals/2

26. At which pH are ammonium biurate crystals usually found in urine?

- A. Acid urine only
- B. Acid or neutral urine
- C. Neutral or alkaline urine
- D. Alkaline urine only

Body fluids/Correlate laboratory data with physiological processes/Urine crystals/2

27. Which of the following crystals is seen commonly in alkaline and neutral urine?

- A. Calcium oxalate
- B. Uric acid
- C. Magnesium ammonium phosphate
- D. Cholesterol

Body fluids/Correlate laboratory data with physiological processes/Urine crystals/1

28. Which crystal appears in urine as a long, thin hexagonal plate and is linked to ingestion of large amounts of benzoic acid?

- A. Cystine
- B. Hippuric acid
- C. Oxalic acid
- D. Uric acid

Body fluids/Correlate laboratory data with physiological processes/Urine crystals/2

23. **B** Flat, six-sided uric acid crystals may be mistaken for cystine crystals. Both crystals form at an acid to neutral pH. Cystine crystals are colorless, whereas uric acid crystals are pigmented (yellow, reddish brown). Cystine transmits polarized light and is soluble in dilute HCl. Uric acid is insoluble in HCl and is less anisotropic. Cystine is reduced by sodium cyanide, forming cysteine. The -SH group of cysteine reacts with nitroprusside to form a red color.
24. **B** Tyrosine crystals may occur in tyrosinemia, an inborn error of tyrosine metabolism caused by a deficiency of fumarylacetoacetate hydrolase, p-hydroxyphenylpyruvic acid oxidase, or tyrosine aminotransferase (causes of tyrosinuria). However, when seen along with leucine crystals, the cause is chronic liver disease, usually cirrhosis of the liver. Tyrosine usually forms fine brown or yellow needles, and leucine forms yellow spheres with concentric rings.
25. **C** Abnormal crystals are those that result from a pathological process. Hemosiderin crystals result from intravascular RBC destruction. Bilirubin crystals are found in severe necrotic and obstructive liver diseases, and cholesterol crystals in nephrotic syndrome, diabetes mellitus, and hypercholesterolemia.
26. **D** Ammonium biurate is often called a “thornapple” crystal because it forms a dark brown spiny sphere. Calcium carbonate (CaCO_3) is another common crystal that is seen only in alkaline urine. Sodium urate and uric acid form in acid or neutral urine.
27. **C** Magnesium ammonium phosphate, also called *triple phosphate*, may be present in neutral or alkaline urine. Most commonly, triple phosphate crystals are six-sided plates that resemble a coffin lid. Crystals containing phosphates do not occur in acid urine.
28. **B** Hippuric acid forms long, colorless, flat, six-sided plates. It results from the metabolism of benzoic acid and resembles the “coffin lid” appearance of triple phosphate. It may occur normally as a result of ingestion of vegetables preserved with benzoic acid.

29. Small, yellow needles are seen in the sediment of a urine sample with a pH of 6.0. Which of the following crystals can be ruled out?

- A. Sulfa crystals
- B. Bilirubin crystals
- C. Uric acid crystals
- D. Cholesterol crystals

Body fluids/Apply knowledge of fundamental biological characteristics/Urine crystals/2

30. Oval fat bodies are derived from:

- A. Renal tubular epithelium
- B. Transitional epithelium
- C. Degenerated WBCs
- D. Mucoprotein matrix

Body fluids/Apply knowledge of fundamental biological characteristics/Urine sediment/1

31. Oval fat bodies are often associated with:

- A. Lipoid nephrosis
- B. Acute glomerulonephritis
- C. Aminoaciduria

D. Pyelonephritis

Body fluids/Correlate clinical and laboratory data/Urine sediment/2

32. Urine of constant SG ranging from 1.008 to 1.010 most likely indicates:

- A.** Addison disease
- B.** Renal tubular failure
- C.** Prerenal failure
- D.** Diabetes insipidus

Body fluids/Evaluate laboratory data to recognize health and disease states/Specific gravity/2

33. Which of the following characterizes prerenal failure, and helps to differentiate it from acute renal failure caused by renal disease?

- A.** BUN:creatinine ratio of 20:1 or higher
- B.** Urine:plasma osmolal ratio less than 2:1
- C.** Excess loss of sodium in urine
- D.** Dehydration

Body fluids/Correlate clinical and laboratory data/Renal disease/2

34. Which of the following conditions characterizes chronic glomerulonephritis and helps differentiate it from acute glomerulonephritis?

- A.** Hematuria
- B.** Polyuria
- C.** Hypertension
- D.** Azotemia

Body fluids/Correlate clinical and laboratory data/Renal disease/2

Answers to Questions 29–34

- 29. D** Cholesterol crystals are colorless rectangular plates that often have a notched corner and appear stacked in a stair-step arrangement. Cholesterol crystals are highly anisotropic and can be positively identified by using a polarizing microscope. Bilirubin, sulfa, or uric acid crystals may occur as small yellow or yellow-brown needles or rods in neutral or acid urine. Bilirubin crystals should be suspected when the dry reagent strip test for bilirubin yields a positive result and cells in the sediment are dark yellow (bile stained). Sulfa crystals are soluble in acetone, concentrated HCl, and NaOH.
- 30. A** Oval fat bodies form from degenerated renal epithelial cells that have reabsorbed cholesterol from the filtrate. They stain with Oil Red O or Sudan III. The fat globules within the cells give a Maltese cross effect when examined under polarized light.
- 31. A** The term *lipoid nephrosis* is a synonym for idiopathic (primary) nephrotic syndrome. Like other forms of nephrotic syndrome, it is associated with gross proteinuria, edema, and hyperlipidemia; however, the idiopathic form is also associated with hematuria. It is common in children and is often associated with mutations that affect podocyte structure.
- 32. B** The SG of the filtrate in the Bowman space is approximately 1.010. Urine produced consistently with a SG of 1.010 has the same osmolality of the plasma and results from

failure of the tubules to modify the filtrate.

- 33. A** Prerenal failure is caused by deficient renal blood flow. The tubules are undamaged and will reabsorb more BUN than normal because filtrate flow is slow. Under the influence of aldosterone, they reabsorb sodium and concentrate the urine. The BUN:creatinine ratio and the U:P osmolal ratio are very high, and sodium output is low. In renal disease, the BUN:creatinine ratio is 10 or less, the U:P osmolal ratio approaches 1.0, and the daily sodium excretion is high.
- 34. B** Acute glomerulonephritis results in severe compression of the glomerular vessels. This reduces filtration, causing a progression from oliguria to anuria. In contrast, polyuria is associated with chronic glomerulonephritis, which causes scarring of the collecting tubules. Both acute and chronic glomerulonephritis cause low urine osmolality, azotemia, acidosis, hypertension, proteinuria, and hematuria.

35. Which of the following conditions is seen in acute renal failure and helps differentiate it from prerenal failure?

- A. Hyperkalemia and uremia
- B. Oliguria and edema
- C. Low creatinine clearance
- D. Abnormal urinary sediment

Body fluids/Correlate clinical and laboratory data/Renal disease/2

36. Which of the following conditions characterizes acute renal failure and helps differentiate it from chronic renal failure?

- A. Hyperkalemia
- B. Hematuria
- C. Cylindruria
- D. Proteinuria

Body fluids/Correlate clinical and laboratory data/Renal disease/2

37. The serum concentration of which analyte is likely to be decreased in untreated cases of acute renal failure?

- A. Hydrogen ions
- B. Inorganic phosphorus
- C. Calcium
- D. Uric acid

Body fluids/Correlate clinical and laboratory data/Renal disease/2

38. Which of the following conditions is associated with the greatest proteinuria?

- A. Acute glomerulonephritis
- B. Chronic glomerulonephritis
- C. Nephrotic syndrome
- D. Acute pyelonephritis

Body fluids/Correlate clinical and laboratory data/Renal disease/2

39. Which of the following conditions is often a cause of glomerulonephritis?

- A. Hypertension
- B. CMV infection
- C. Systemic lupus erythematosus (SLE)
- D. Heavy metal poisoning

Answers to Questions 35–39

- 35. D** Reduced glomerular filtration as evidenced by low creatinine clearance characterizes both prerenal failure and acute renal failure. This results in retention of fluid, causing edema, reduced urine volume, hypertension, uremia, and hyperkalemia in both prerenal failure and acute renal failure. The kidneys are not damaged in prerenal failure, and therefore, the microscopic examination is usually normal.
- 36. A** In acute renal failure, reduced glomerular filtration coupled with decreased tubular secretion results in hyperkalemia. In chronic renal failure, scarring of the collecting tubules prevents salt and H₂O reabsorption. This can result in normal or low serum potassium despite reduced glomerular filtration. The sediment in chronic renal failure is characterized by intermittent heavy hematuria and proteinuria.
- 37. C** Decreased glomerular filtration in renal failure results in high serum creatinine, BUN, and uric acid. Failure of the tubules results in retention of hydrogen ions and phosphates, causing acidosis and an increased anion gap. The tubules fail to respond to parathyroid hormone, resulting in excessive loss of calcium in urine. Serum sodium is usually normal or slightly increased, whereas hyperkalemia is a constant finding in acute renal failure.
- 38. C** Although all four conditions are associated with proteinuria, it is greatest in nephrotic syndrome. Urinary albumin loss is typically in excess of 4 g/day or 3 mg/mg creatinine, causing dry reagent strip protein tests to give 3+ to 4+ reactions. In contrast to glomerulonephritis and pyelonephritis, urinary sediment in nephrotic syndrome in adults is not usually characterized by either hematuria or pyuria. Various casts, lipid laden renal epithelial cells, and oval fat bodies are usually found.
- 39. C** Autoimmune diseases, diabetes mellitus, and nephrotoxic drugs are common causes of acute glomerulonephritis. Autoimmune damage may result from the deposition of antigen–antibody complexes and complement-mediated damage, as occurs in poststreptococcal glomerulonephritis, or from the production of autoantibodies that attack the basement membrane as in Goodpasture syndrome. Acute glomerulonephritis is often classified by the pattern of injury rather than by the cause. For example, insulin deficiency produces sclerotic vascular damage to the glomeruli, often resulting in crescentic glomerulonephritis. Group A *Streptococcus* and SLE result in immunologically mediated damage to the glomeruli usually causing membranous or membranoproliferative glomerulonephritis. CMV infections and heavy metal poisoning cause damage to the tubules, resulting in nephrosis.

40. Acute pyelonephritis is commonly caused by:

- A.** Bacterial infection of medullary interstitium
- B.** Circulatory failure
- C.** Renal calculi
- D.** Antigen–antibody reactions within the glomeruli

41. Which of the following is associated with nephrotic syndrome?

- A. Hyperlipidemia
- B. Uremia
- C. Hematuria and pyuria
- D. Dehydration

Body fluids/Correlate clinical and laboratory data/Renal disease/2

42. Which of the following conditions is a characteristic finding in patients with obstructive renal disease?

- A. Polyuria
- B. Azotemia
- C. Dehydration
- D. Alkalosis

Body fluids/Correlate laboratory data with physiological processes/Renal disease/2

43. Whewellite and weddellite kidney stones are composed of:

- A. Magnesium ammonium phosphate
- B. Calcium oxalate
- C. Calcium phosphate
- D. Calcium carbonate

Body fluids/Apply knowledge of fundamental biological characteristics/Renal calculi/1

Answers to Questions 40–43

- 40. A** Acute pyelonephritis is caused by infection of the medullary interstitium, usually by coliforms that enter from the lower urinary tract. *Escherichia coli* is the most commonly implicated bacterium. Because it is focused in the medulla, the disease involves mainly the tubules. As opposed to acute glomerulonephritis, pyelonephritis is not associated with reduced creatinine clearance, azotemia, or oliguria. Reabsorption of salt and water are blocked, resulting in hyperkalemia, acidosis, and polyuria.
- 41. A** Although casts may be present, the urinary sediment in nephrotic syndrome in adults is not characterized by RBCs and WBCs or by RBC, blood, and WBC casts. In nephrotic syndrome, unlike renal failure (nephritic syndrome), the creatinine clearance and serum potassium are usually normal. Nephrotic syndrome can be secondary to other renal diseases, infections, and drug treatment. In such cases, it is often transient. When it follows the anuric phase of acute glomerulonephritis, it signals a reversal in the course of the disease.
- 42. B** Obstructive renal disease may result from renal or urinary tract calculi, benign prostatic hypertrophy, chronic urinary tract infection, or urogenital malignancy. Obstruction causes the hydrostatic pressure in the Bowman space to increase. This pressure opposes glomerular filtration. If the hydrostatic pressure in the Bowman space equals the hydrostatic pressure in the glomeruli, then filtration will not occur, resulting in anuria. Postrenal failure produces many of the same serum abnormalities as acute renal failure, including hyperkalemia, acidosis, edema, and azotemia. Urinary sediment will often be abnormal, as well. Bacteriuria and pyuria are common, and hematuria may result from rupture of the vasa recta or other blood vessels.
- 43. B** Over three-fourths of urinary tract stones are composed of calcium salts, and

hyperparathyroidism is commonly associated with calcium stones. Stones composed of magnesium ammonium phosphate are called *struvite* and lodge in the renal pelvis, causing a characteristic “staghorn” appearance on radiographic examination. Stones mainly composed of calcium phosphate are called *hydroxyapatite* or *bushite*, depending on the calcium composition. Stones of CaCO_3 are called *carbonate apatite*.

44. Which of the following abnormal crystals is often associated with formation of renal calculi?

- A. Cystine
- B. Ampicillin
- C. Tyrosine
- D. Leucine

Body fluids/Correlate clinical and laboratory data/Renal calculi/2

45. Which statement about renal calculi is true?

- A. Calcium oxalate and calcium phosphate account for about three-fourths of all stones
- B. Uric acid stones can be seen on radiography
- C. Triple phosphate stones are found principally in the ureters
- D. Stones are usually composed of single salts

Body fluids/Apply knowledge of fundamental biological characteristics/Renal calculi/2

Answers to Questions 44–45

- 44. A** Cystinuria is caused by an autosomal recessive defect in the tubular reabsorption of dibasic amino acids (a renal-type aminoaciduria). Cystine crystals are highly insoluble and form kidney stones. Tyrosine crystals form fine dark sheaves or needles and may result from liver disease or tyrosinosis, an overflow aminoaciduria. Leucine crystals form yellow spheres with concentric rings and are seen in chronic liver disease. Ampicillin (rarely) forms long colorless prisms in sheaves in some patients being treated with high doses.
- 45. A** Three-fourths of all stones contain calcium, and three-fourths of these contain calcium oxalate. Stones are usually composed of several inorganic salts, but calcium oxalate is the most common component of urinary stones. Oxalates are hard, dark, and coarse stones. Uric acid stones are always pigmented yellow to reddish brown. They are small translucent stones not apparent on radiography. Stones made of primarily calcium phosphate (as hydroxyapatite) are light and crumble easily. Stones made of struvite (ammonium magnesium phosphate) are radiodense and lodge in the renal pelvis, forming an outline of the structures resembling the antlers of a deer (staghorn calculi).

7.3

Cerebrospinal, Serous, and Synovial Fluids

1. Cerebrospinal fluid (CSF) is formed by ultrafiltration of plasma through the:

- A. Choroid plexus
- B. Sagittal sinus
- C. Anterior cerebral lymphatics
- D. Arachnoid membrane

Body fluids/Apply knowledge of fundamental biological characteristics/Cerebrospinal fluid/1

2. Which statement regarding CSF is true?

- A. Normal values for mononuclear cells are higher for infants than for adults
- B. Absolute neutrophilia is not significant if the total WBC count is less than 25/ μ L
- C. The first aliquot of CSF should be sent to the microbiology laboratory
- D. Neutrophils compose the majority of WBCs in normal CSF

Body fluids/Apply principles of basic laboratory procedures/Cerebrospinal fluid/2

3. When collecting CSF, a difference between opening and closing fluid pressure greater than 100 mm H₂O indicates:

- A. Low CSF volume
- B. Subarachnoid hemorrhage
- C. Meningitis
- D. Hydrocephalus

Body fluids/Correlate laboratory data with physiological processes/Cerebrospinal fluid/2

Answers to Questions 1–3

- 1. A** CSF is formed by ultrafiltration of plasma through the choroid plexus, a tuft of capillaries in the pia mater located in the third and fourth ventricles. Endothelia of the choroid plexus vessels and ependymal cells lining the ventricles act as a barrier to the passage of proteins, drugs, and metabolites. Glucose in CSF is about 60% of the plasma glucose. Total protein in CSF is only 15 to 45 mg/dL, while chloride levels are 10% to 15% higher than plasma. Approximately 500 mL of ultrafiltrate is produced per day, the bulk of which is returned to the circulation via the sagittal sinus. The normal volume of CSF in adults is 100 to 160 mL (10–60 mL for small children).
- 2. A** Lymphocytes account for 40% to 80% of WBCs in adults; monocytes and macrophages for 20% to 50%. Neutrophils should be less than 10% of the WBCs. The reference range for WBCs in adults is 0 to 5/ μ L. A majority of PMNs can be significant even if the WBC count is normal. In infants, monocytes account for 50% to 90% of WBCs, and the upper limit for WBCs is 30/ μ L. The first aliquot is sent to the chemistry department because it may be contaminated with blood or skin flora.
- 3. A** Normal CSF volume in adults is 100 to 160 mL. When volume is low, an abnormally

high difference is observed between the opening and closing pressure. The difference is normally 10 to 30 mm H₂O, after removal of 15 to 20 mL. Low opening pressure is caused by reduced volume or block above the puncture site. High opening pressure may result from high CSF volume, CNS hemorrhage, or malignancy.

4. Which of the following findings is consistent with a subarachnoid hemorrhage rather than a traumatic tap?

- A. Clearing of the fluid as it is aspirated
- B. A clear supernatant after centrifugation
- C. Xanthochromia
- D. Presence of a protein in the sample

Body fluids/Evaluate laboratory data to recognize health and disease states/Cerebrospinal fluid/2

5. The term used to denote a high WBC count in the CSF is:

- A. Empyema
- B. Neutrophilia
- C. Pleocytosis
- D. Hyperglycorrhachia

Body fluids/Apply knowledge of fundamental biological characteristics/Cerebrospinal fluid/1

Answers to Questions 4–5

- 4. **C** Xanthochromia is pigmentation of CSF caused by subarachnoid hemorrhage, high CSF protein, free hemoglobin, or bilirubin. The bilirubin may be caused by hepatic disease, CNS hemorrhage, or prior traumatic tap. In subarachnoid hemorrhage, the fluid will be pink if the RBC count is greater than 500/μL. It will turn orange as RBCs lyse in the first few hours and then will turn yellow after about 12 hours. Granulocyte infiltration occurs immediately after a subarachnoid hemorrhage and disappears after 24 hours. It is followed by an increase in macrophages, showing evidence of erythrophagocytosis that remains for up to 2 weeks. After subarachnoid hemorrhage, D-dimer is present in CSF and can be used to distinguish between a traumatic tap and subarachnoid hemorrhage.
- 5. **C** Pleocytosis refers to an increase in WBCs within CSF. Bacterial meningitis causes a neutrophilic pleocytosis, viral meningitis a lymphocytic pleocytosis, and tuberculous and fungal meningitis a mixed-cell pleocytosis. Other causes of pleocytosis include multiple sclerosis (MS), cerebral hemorrhage or infarction, and leukemia.

6. Which of the adult CSF values in the table below are consistent with bacterial meningitis?

	WBCs	Lymphocytes	Monocytes	Eosinophils	Neutrophils	Neuroectodermal Cells
A.	50/μL	44%	55%	0%	0%	1%
B.	300/μL	75%	21%	3%	0%	1%
C.	2,000/	5%	15%	0%	80%	0%

	μL					
D.	2,500/ μL	40%	50%	0%	10%	0%

Body fluids/Evaluate laboratory data to recognize health and disease states/Cerebrospinal fluid/2

7. Given the following data, determine the corrected CSF WBC count.

	CSF Values	Peripheral Blood Values
RBCs	6,000/μL	$4.0 \times 10^6/\mu\text{L}$
WBCs	150/μL	$5.0 \times 10^3/\mu\text{L}$

- A. 8/μL
- B. 142/μL
- C. 120/μL
- D. 145/μL

Body fluids/Calculate/CSF hematology/2

8. **SITUATION:** What is the *most likely* cause of the following CSF results?

CSF glucose 20 mg/dL; CSF protein 200 mg/dL;
CSF lactate 50 mg/dL (reference range 5–25 mg/dL)

- A. Viral meningitis
- B. Viral encephalitis
- C. Cryptococcal meningitis
- D. Acute bacterial meningitis

Body fluids/Evaluate laboratory data to recognize health and disease states/Cerebrospinal fluid/2

9. Which of the following conditions is *most often* associated with normal CSF glucose and protein?

- A. Multiple sclerosis
- B. Malignancy
- C. Subarachnoid hemorrhage
- D. Viral meningitis

Body fluids/Correlate clinical and laboratory data/Cerebrospinal fluid/2

Answers to Questions 6–9

6. **C** Normal WBC counts for CSF are 0 to 5/μL for adults and 0 to 30/μL for children. Neutrophils predominate the differential in bacterial meningitis, whereas lymphocytes predominate in viral meningitis. Hemorrhage and traumatic tap will also cause increased PMNs, and WBC counts should be corrected using the CSF RBC count.
7. **B** Corrected WBC count = WBCs in CSF – [(Blood WBCs × CSF RBCs) ÷ Blood RBCs]
 Corrected WBC count = 150/μL – [(5,000/μL WBCs × 6,000/μL RBCs) ÷ 4,000,000/μL RBCs]
 Corrected WBC count = 150/μL – 7.5/μL

Corrected WBC count = 142/ μ L

- 8. D** Acute bacterial meningitis causes increased production of immunoglobulin (Ig) in CSF. Glucose levels are below normal (less than 40 mg/dL) as a result of consumption by PMNs and bacteria. Lactate levels rise because of increased pressure and hypoxia (greater than 35 mg/dL being correlated with bacterial meningitis). When associated with increased PMNs and lactate dehydrogenase (LD), these findings point to bacterial meningitis.
- 9. D** In viral (aseptic) meningitis, the CSF glucose is usually above 40 mg/dL and the total protein is normal or slightly increased. Some types of viral meningitis can cause low glucose and high protein levels, making the differentiation of bacterial and viral meningitis difficult. In such cases, an elevated CSF C-reactive protein (CRP) and procalcitonin (PCT) strongly support the diagnosis of bacterial meningitis. PCT is increased in response to inflammation. Like lactate and CRP, it is useful for diagnosis of septic shock and following the response to antibiotic therapy. Low CSF glucose and elevated total protein are also seen in malignancy and subarachnoid hemorrhage and in some persons with MS. Low glucose in malignancy and MS results from increased utilization. Glucose is reduced in subarachnoid hemorrhage as a result of release of glycolytic enzymes from the RBCs. All three conditions result in high CSF protein, but MS is associated with an increased IgG index owing to local production of IgG.

10. The diagnosis of MS is suggested by which finding?

- A. The presence of elevated protein and low glucose
- B. A decreased IgG index
- C. The presence of oligoclonal bands by electrophoresis
- D. An increased level of CSF β -microglobulin

Body fluids/Correlate clinical and laboratory data/Cerebrospinal fluid/2

11. Which of the following results is consistent with fungal meningitis?

- A. Normal CSF glucose
- B. Pleocytosis of mixed cellularity
- C. Normal CSF protein
- D. High CSF lactate

Body fluids/Correlate clinical and laboratory data/Cerebrospinal fluid/2

12. In what suspected condition should a wet prep using a warm slide be examined?

- A. Cryptococcal meningitis
- B. Amoebic meningoencephalitis
- C. *Mycobacterium tuberculosis* infection
- D. Neurosyphilis

Body fluids/Select course of action/Cerebrospinal fluid/2

13. Which of the following CSF test results is *most* commonly increased in patients with MS?

- A. Glutamine
- B. Lactate
- C. IgG index
- D. Ammonia

Body fluids/Correlate clinical and laboratory data/Cerebrospinal fluid/2

Answers to Questions 10–13

- 10. C** The total CSF protein is increased in less than half of persons with MS. The IgG index is increased in 80% or more of MS cases. Although the IgG index is sensitive, it is increased in many other disorders. The presence of oligoclonal banding (two or more discrete bands in the gamma zone following electrophoresis) is seen in 90% of persons with MS, and in few other diseases. Although not entirely definitive, it is the single most effective laboratory test for the diagnosis of MS. When performing CSF electrophoresis, the serum pattern must be compared with the CSF pattern. At least some of the oligoclonal bands must *not* be found in the serum pattern for the test result to be considered positive. β_2 -Microglobulins are increased in CSF in inflammatory diseases (especially malignant diseases).
- 11. B** In fungal meningitis, the glucose is low and the total protein elevated; however, unlike bacterial meningitis, the lactate is usually below 35 mg/dL. Fungal meningitis usually produces a pleocytosis of mixed cellularity consisting of lymphocytes, PMNs, monocytes, and eosinophils. In some cases, lymphocytes predominate, whereas in others, PMNs comprise the majority of WBCs.
- 12. B** Amoeba in CSF appear very similar to monocytes in stained films but can be differentiated by their characteristic pseudopod mobility in a wet prep on a prewarmed slide. *Naegleria fowleri* and *Acanthamoeba* spp. are causative agents of primary amoebic meningoencephalitis.
- 13. C**
$$\text{IgG Index} = \frac{(\text{CSF IgG} \div \text{serum IgG})}{(\text{CSF albumin} \div \text{serum albumin})}$$
 An IgG–albumin index is the ratio of CSF IgG:serum IgG divided by the CSF albumin:serum albumin ratio. Values greater than 0.85 indicate CSF IgG production, as seen in MS; or increased CSF production combined with increased permeability, as seen in CNS infections. MS is characterized by the presence of oligoclonal banding in the CSF in greater than 90% of patients with active disease. The total protein and myelin basic protein are often increased, and glucose is decreased. Reye syndrome results in hepatic failure, causing high CSF levels of ammonia and glutamine. CSF lactate is usually normal in patients with MS.
- 14. Which of the following is an *inappropriate* procedure for performing routine CSF analysis?**
- A. A differential is done only if the total WBC count is greater than 10/ μ L
 - B. A differential should be done on a stained CSF concentrate
 - C. A minimum of 30 WBCs should be differentiated
 - D. A Wright-stained slide should be examined, rather than a chamber differential
- Body fluids/Apply principles of standard operating procedures/Cerebrospinal fluid/2*
- 15. Which cell is present in CSF in greater numbers in newborns than in older children or adults?**
- A. Eosinophils
 - B. Lymphocytes
 - C. Monocytes
 - D. Neutrophils

Body fluids/Correlate clinical and laboratory data/Cerebrospinal fluid/2

16. Neutrophilic pleocytosis is usually associated with all of the following except:

- A. Cerebral infarction
- B. Malignancy
- C. Myelography
- D. Neurosyphilis

Body fluids/Correlate clinical and laboratory data/Cerebrospinal fluid/2

17. Which statement about CSF protein is true?

- A. An abnormal serum protein electrophoretic pattern does not affect the CSF pattern
- B. The upper reference limit (URL) for CSF total protein in newborns is one half the adult level
- C. CSF IgG is increased in panencephalitis, malignancy, and neurosyphilis
- D. Antibodies to *Treponema pallidum* disappear after successful antibiotic therapy

Body fluids/Correlate clinical and laboratory data/Cerebrospinal fluid/2

18. Which of the following statements regarding routine microbiological examination of CSF is true?

- A. A Gram stain is performed on the CSF prior to concentration
- B. The Gram stain is positive in fewer than 40% of cases of acute bacterial meningitis
- C. India ink and acid fast stains are indicated if neutrophilic pleocytosis is present
- D. All CSF specimens should be cultured using sheep blood agar, chocolate agar, and supplemented broth

Body fluids/Apply knowledge of standard operating procedures/Cerebrospinal fluid/2

Answers to Questions 14–18

- 14. A** A relative (percent) increase in PMNs may be significant even when the WBC count does not exceed the upper limit of normal. For this reason, a WBC differential using a concentrated CSF sample is always performed on neonates and when the WBC count is greater than 5/ μ L. Cytocentrifugation should be used to concentrate the cells followed by staining with the Wright stain.
- 15. C** In newborns, the URL for WBCs is 30/ μ L (URL for adults is 5/ μ L) with the majority of WBCs being monocytes or macrophages. In normal neonates, monocytes (including macrophages and histiocytes) account for about 75% of the WBCs, lymphocytes for about 20%, and PMNs for about 3%. In normal adults, lymphocytes account for about 60% of the WBCs, monocytes for about 35%, and PMNs for about 2%.
- 16. D** Neutrophils may appear in CSF as a result of many causes, making it necessary to correlate results of biochemical assays with hematologic findings. Low glucose and high protein levels occur in both malignancy and bacterial meningitis. Tumor markers and lactate may be helpful in distinguishing malignancy from bacterial meningitis. In neurosyphilis, there is usually an absolute lymphocytosis and increased total protein and IgG index.
- 17. C** Although the blood–brain barrier excludes most plasma proteins, abnormal serum proteins can cause parallel CSF electrophoretic patterns. Therefore, an abnormal CSF pattern indicates CNS disease only if not duplicated by the serum pattern. Normal CSF

total protein in newborns may be up to two times higher than adult levels. Antibodies to *T. pallidum* remain in CSF after treatment, but nontreponemal antibodies disappear. Although the *T. pallidum* particle agglutination (TP-PA) test for specific antibodies is more sensitive, the Venereal Disease Research Laboratory (VDRL) test is often performed concurrently. A positive result for both tests is diagnostic of active tertiary syphilis.

- 18. D** A culture should be performed on the sediment of the third aliquot of CSF after it is centrifuged. Blood and chocolate agar and anaerobic broth should always be used, and, if sterile, held for a minimum of 3 days. Blood cultures should be done because septicemia occurs in about one half of bacterial meningitis cases. Gram staining is always performed by using the sediment of the CSF because it is positive in more than 70% of acute bacterial meningitis cases. India ink, acid-fast, and wet preparations may be ordered if an absolute monocytosis is present.

- 19. Which organism is the most frequent cause of bacterial meningitis in neonates?**

- A. *Neisseria meningitidis*
- B. Group B *Streptococcus*
- C. *Haemophilus influenzae*
- D. *Klebsiella pneumoniae*

Body fluids/Correlate clinical and laboratory data/Cerebrospinal fluid/2

- 20. Following a head injury, which protein will identify the presence of CSF leakage through the nose?**

- A. Transthyretin
- B. Myelin basic protein
- C. Tau protein
- D. C-reactive protein

Body fluids/Select test/CSF/2

- 21. Which of the following statements regarding serous fluids is true?**

- A. The normal volume of pleural fluid is 30 to 50 mL
- B. Mesothelial cells, PMNs, lymphocytes, and macrophages may be present in normal fluids
- C. Radiography can detect a 10% increase in the volume of a serous fluid
- D. Normal serous fluids are colorless

Body fluids/Correlate clinical and laboratory data/Serous fluid/2

- 22. The term *effusion* refers to:**

- A. A chest fluid that is purulent
- B. A serous fluid that is chylous
- C. An increased volume of serous fluid
- D. An inflammatory process affecting the appearance of a serous fluid

Body fluids/Apply knowledge of fundamental biological characteristics/Pleural fluid/1

- 23. Which of the following laboratory results is characteristic of a transudative fluid?**

- A. SG = 1.018
- B. Total protein = 3.2 g/dL
- C. LD fluid/serum ratio = 0.25
- D. Total protein fluid:serum ratio = 0.65

Body fluids/Evaluate laboratory data to recognize health and disease states/Exudates/2

Answers to Questions 19–23

19. **B** Group B *Streptococcus* and *E. coli* are the two most common isolates in neonates. *H. influenzae*, *S. pneumoniae*, and *N. meningitidis* are the most common isolates in children. *S. pneumoniae* is the most frequent isolate in older adults.
20. **C** In cases of trauma, it may be necessary to differentiate rhinorrhea from CSF leakage, and this can be done by immunofixation electrophoresis to identify the tau protein found in CSF, but not in serum. Tau protein is an enzymatically modified form of transferrin that migrates in the slow beta zone just behind unmodified transferrin. Transthyretin or prealbumin is present in far greater concentration in CSF than in blood but may not be seen if CSF is diluted with nasal fluid. Myelin basic protein is a component of nerve sheaths and is present in CSF in about 60% of persons with MS. It is also found in persons with other demyelinating diseases, SLE, stroke, and brain injury. CRP is elevated in the CSF of approximately two thirds of persons with bacterial meningitis.
21. **B** Serous fluids include pleural, pericardial, and peritoneal fluids. They form from ultrafiltration of plasma through serous membranes. These are lined with specialized epithelium called *mesothelium*. They comprise about 5% of the cells in serous fluid and may be difficult to differentiate from malignant cells. Pleural fluid volume is normally less than 10 mL. The volume of pericardial fluid is normally 10 to 50 mL and peritoneal fluid 30 to 50 mL. Radiography can detect an increase in serous fluids of 300 mL or more. Normal serous fluids are clear and range in color from straw to light yellow.
22. **C** Effusions are classified as either transudates, exudates, or chylous. Transudates result from abnormal hemodynamics (e.g., congestive heart failure, liver disease), and exudates and chylous fluids from local disease. Pleural fluid that is purulent is called *empyemic fluid*. Such fluid has a WBC count of 10,000/ μ L or greater.
23. **C** Transudative fluids are distinguished from exudative fluids by the physical appearance, cellularity, SG, total protein, LD, cholesterol, and bilirubin. Exudative fluids have a fluid:serum LD ratio greater than 0.6 caused by release of the enzyme from inflammatory or malignant cells. Exudative fluids have a total protein greater than 3.0 g/dL, SG greater than 1.015, fluid:serum total protein ratio greater than 0.6, cholesterol greater than 60 mg/dL (fluid:serum ratio greater than 0.3) and fluid:serum bilirubin ratio greater than 0.6. Exudates are caused by infection, infarction, malignancy, rheumatoid diseases, and trauma.

24. Which observation is *least* useful in distinguishing a hemorrhagic serous fluid from a traumatic tap?

- A. Clearing of fluid as it is aspirated
- B. Presence of xanthochromia
- C. The formation of a clot
- D. Diminished RBC count in successive aliquots

Body fluids/Correlate laboratory data with physiological processes/Serous fluids/2

25. Which of the following laboratory results on a serous fluid is *most* likely to be caused by

a traumatic tap?

- A.** An RBC count of 8,000/ μ L
- B.** A WBC count of 6,000/ μ L
- C.** A hematocrit of 35%
- D.** A neutrophil count of 55%

Body fluids/Correlate laboratory data with physiological processes/Serous fluid/2

26. Which of the following conditions is commonly associated with an exudative effusion?

- A.** Congestive heart failure
- B.** Malignancy
- C.** Nephrotic syndrome
- D.** Cirrhosis

Body fluids/Correlate clinical and laboratory data/Transudate/2

27. Which of the following conditions is associated with a chylous effusion?

- A.** Necrosis
- B.** Pulmonary infarction or infection
- C.** Systemic lupus erythematosus or rheumatoid arthritis (RA)
- D.** Lymphatic obstruction

Body fluids/Correlate clinical and laboratory data/Exudates/2

28. Which of the following conditions is most often associated with a pleural fluid glucose below 30 mg/dL?

- A.** Diabetes mellitus
- B.** Pancreatitis
- C.** Rheumatoid arthritis
- D.** Bacterial pneumonia

Body fluids/Correlate clinical and laboratory data/Pleural fluid/2

Answers to Questions 24–28

- 24. C** Xanthochromia indicates either an exudative process or prior traumatic tap. Hemorrhagic pleural fluids usually have RBC counts greater than 100,000/ μ L and are usually caused by lung neoplasms. Clearing of fluid or diminished RBC counts in successive tubes favors a diagnosis of a traumatic tap. A clot may form in a hemorrhagic fluid or following a traumatic tap. However, a transudative fluid will not clot.
- 25. A** Normal fluids have a WBC count less than 1,000/ μ L, but counts between 1,000 and 2,500/ μ L may be seen in both exudates or transudates. All WBC types are present, but no type should account for greater than 50% of the leukocyte count. An RBC count less than 10,000/ μ L is usually caused by a traumatic tap. A fluid hematocrit similar to blood is caused by a hemothorax. Pleural fluids containing greater than 100,000/ μ L RBCs are associated most often with malignancies but are also seen in trauma and pulmonary infarction.
- 26. B** Transudative fluids are caused by circulatory problems, usually decreased oncotic pressure or increased hydrostatic pressure. In contrast, exudative effusions are caused by inflammatory processes and cellular infiltration as seen in malignancy. In addition to an RBC count greater than 100,000/ μ L, malignancies often involve the lung, colon,

breast, or pancreas and often produce carcinoembryonic antigen.

27. **D** Malignancy, pulmonary infarction, SLE, and RA are characterized by inflammation, with increases in protein, WBCs, and LD. Exudates can also be caused by tuberculosis, pancreatitis, and lymphoma. Lymphatic obstruction is often associated with lymphoma and other malignancies that block the flow of lymph into the azygous vein. This causes chylous effusions. Chylous effusions are also caused by traumatic injury to the thoracic duct. Necrosis causes a pseudo-chylous effusion. This resembles a chylous effusion in appearance but has a foul odor. Chylous fluids contain chylomicrons, stain positive for fat globules, show lymphocytosis, and have a triglyceride concentration over twofold higher than plasma (or greater than 110 mg/dL). Pseudo-chylous effusions are characterized by mixed cellularity and elevated cholesterol.
28. **C** Normal pleural fluid has the same glucose concentration as plasma. Hyperglycemia is the only condition that is associated with a high pleural fluid glucose. Low glucose levels (less than 60 mg/dL) may be seen in infection, malignancy, and rheumatic diseases. However, glucose levels are lowest (often below 30 mg/dL) and are a constant finding when rheumatoid disease affects the lungs. Pancreatitis causes an exudative peritoneal and pleural effusion with an elevated peritoneal fluid amylase (without a low glucose).

29. **In which condition is the pleural fluid pH likely to be above 7.3?**

- A. Bacterial pneumonia with parapneumonic exudate
- B. Rheumatoid pleuritis
- C. Esophageal rupture
- D. Pneumothorax

Body fluids/Correlate clinical and laboratory data/Pleural fluid/2

30. **Which of the following hematology values best frames the URLs for peritoneal fluid?**

	WBC Count	Percentage of PMNs	RBC Count
A.	300/ μ L	25%	100,000/ μ L
B.	10,000/ μ L	50%	500,000/ μ L
C.	50,000/ μ L	50%	500,000/ μ L
D.	100,000/ μ L	75%	1,000,000/ μ L

Body fluids/Apply knowledge of fundamental biological characteristics/Serous fluids/2

31. **Which of the following characteristics is higher for synovial fluid than for the serous fluids?**

- A. Specific gravity
- B. Glucose
- C. Total protein
- D. Viscosity

Body fluids/Apply knowledge of fundamental biological characteristics/Synovial fluid/1

32. **In which type of arthritis is the synovial WBC count likely to be greater than 50,000/ μ L?**

- A. Septic arthritis

- B. Osteoarthritis
- C. Rheumatoid arthritis
- D. Hemorrhagic arthritis

Body fluids/Correlate clinical and laboratory data/Synovial fluid/2

33. What type of cell is a “ragocyte”?

- A. Cartilage cell seen in inflammatory arthritis
- B. A PMN with inclusions formed by immune complexes
- C. A plasma cell seen in RA
- D. A macrophage containing large inclusions

Body fluids/Apply knowledge of fundamental biological characteristics/Morphology/1

Answers to Questions 29–33

- 29. D** The pH of pleural fluid is approximately 7.64, and values below 7.30 are usually associated with a poorer prognosis and usually require drainage. Esophageal rupture produces the lowest pH with values in the range of 6.0 to 6.3. In addition, pleural fluid pH is low in rheumatoid disease involving the lungs and pleura, some malignancies, and SLE. Low pH and glucose in pleural fluid are seen in lung abscesses and exudative bacterial pneumonia (called *parapneumonic effusion*). Pneumothorax results from air entering the pleural space and does not produce a low pH.
- 30. A** Peritoneal fluid normally has a WBC count less than 300/ μ L. Neutrophils should account for no more than 25% of the WBCs. A majority of PMNs indicates bacterial infection of the peritoneum. Lymphocytosis suggests malignancy, tuberculosis, cirrhosis, and lymphatic leakage. Peritoneal fluid amylase is elevated in most cases of acute pancreatitis. Peritonitis is suspected when the fluid LD is greater than 40% of the serum level. In contrast, normal pleural fluid has a WBC count usually less than 1,000/ μ L. Exudative pleural fluids usually have a WBC count greater than 10,000/ μ L, but values tend to overlap noninflammatory fluids. The PMNs should comprise 50% of the WBCs or less, and the RBC count should be less than 100,000/ μ L. Lesser numbers are usually caused by a traumatic tap.
- 31. D** Synovial fluid has approximately the same SG and glucose as plasma and the serous fluids but is far more viscous because of a high content of mucoprotein (hyaluronate) secreted by the synovium. Viscosity is estimated by pulling the fluid from the tip of a syringe or pipet. Normal fluid gives a string longer than 4 cm. Low viscosity indicates inflammation. The total protein of synovial fluid is usually lower than serous fluids, the URL being 2.0 g/dL.
- 32. A** The WBC count is elevated in all types of arthritis but is greatest (50,000–100,000/ μ L) in septic arthritis. Neutrophils comprise less than 25% of WBCs in normal and noninflammatory arthritis but are greater than 50% in inflammatory and septic arthritis. Fluids are diluted in saline because acetic acid causes a mucin clot to form. WBC counts should be performed within 1 hour of collection because the WBC count will diminish over time.
- 33. B** Ragocytes are PMNs containing dark granules composed of immunoglobulins, but they may be seen in gout and septic arthritis as well as RA. LE cells may be seen in

fluid from patients with SLE. Reiter cells, macrophages with ingested globular inclusions, are seen in Reiter syndrome and other inflammatory diseases.

34. Which of the following crystals is the cause of gout?

- A. Uric acid or monosodium urate
- B. Calcium pyrophosphate or apatite
- C. Calcium oxalate
- D. Cholesterol

Body fluids/Apply knowledge of fundamental biological characteristics/Synovial fluid/1

35. Which crystal causes “pseudogout”?

- A. Oxalic acid
- B. Calcium pyrophosphate
- C. Calcium oxalate
- D. Cholesterol

Body fluids/Apply knowledge of fundamental biological characteristics/Synovial fluid/1

36. A synovial fluid sample is examined by using a polarizing microscope with a red compensating filter. Crystals are seen that are yellow when the long axis of the crystal is parallel to the slow vibrating light. When the long axis of the crystal is perpendicular to the slow vibrating light, the crystals appear blue. What type of crystal is present?

- A. Calcium oxalate
- B. Calcium pyrophosphate
- C. Uric acid
- D. Cholesterol

Body fluids/Apply principles of special procedures/Synovial fluid/2

37. In which condition is the synovial fluid glucose most likely to be within normal limits?

- A. Septic arthritis
- B. Inflammatory arthritis
- C. Hemorrhagic arthritis
- D. Gout

Body fluids/Correlate clinical and laboratory data/Synovial fluid/2

38. Which statement about synovial fluid in RA is true?

- A. Synovial:serum IgG is usually 1:2 or higher
- B. Total hemolytic complement is elevated
- C. Ninety percent of RA cases test positive for rheumatoid factor in synovial fluid
- D. Demonstration of rheumatoid factor in joint fluid is diagnostic for RA

Body fluids/Correlate clinical and laboratory data/Synovial fluid/2

Answers to Questions 34–38

- 34. A** Although all of the crystals mentioned can cause crystal-induced arthritis, uric acid and sodium urate crystals cause gout and are seen in about 90% of patients with gout.
- 35. B** Calcium pyrophosphate crystals occur as needles or small rhombic plates and can be confused with uric acid. They rotate plane polarized light but not as strongly as uric acid. Synovial fluid should never be collected in tubes containing powdered EDTA because it may form crystals that can be mistaken for in vivo crystals. The

recommended anticoagulant is sodium heparin, although liquid EDTA may be used.

- 36. C** Polarized microscopy with a red compensating filter differentiates uric acid from pseudogout crystals. When the long axis of uric acid needles is parallel to the slow vibrating light, the crystals appear yellow. When the long axis is perpendicular to the slow vibrating light, the crystals appear blue. Calcium pyrophosphate gives the reverse effect.
- 37. C** Synovial fluid glucose is normally less than 10 mg/dL below serum glucose, and should be collected after an 8-hour fast to ensure that the fluid and plasma are equilibrated. In septic arthritis, the glucose level is often greater than 40 mg/dL below the serum level and about 25 to 40 mg/dL lower in inflammatory arthritis, which includes gout. Osteoarthritis and hemorrhagic arthritis are not usually associated with low joint fluid glucose.
- 38. A** Rheumatoid factor can be present in both serum and synovial fluids from patients with RA, SLE, and other inflammatory diseases. Rheumatoid factor is present in synovial fluid of approximately 60% of patients with RA. Normally, IgG in synovial fluid is about 10% of the serum IgG level. CH₅₀ levels in serum and synovium are more differential. Both are increased in Reiter syndrome but are often low in SLE; synovial CH₅₀ is decreased and serum CH₅₀ is normal (or increased) in RA.

39. Which of the following organisms accounts for the majority of septic arthritis cases in young and middle-age adults?

- A. *H. influenzae*
B. *Neisseria gonorrhoeae*
C. *Staphylococcus aureus*
D. *Borrelia burgdorferi*

Body fluids/Apply knowledge of fundamental biological characteristics/Synovial fluid/2

40. Which of the following hematology values best frames the URLs for synovial fluid?

	WBC Count	Percentage of PMNs	RBC Count
A.	200/ μ L	25%	2,000/ μ L
B.	5,000/ μ L	50%	10,000/ μ L
C.	10,000/ μ L	50%	50,000/ μ L
D.	20,000/ μ L	5%	500,000/ μ L

Body fluids/Apply knowledge of fundamental biological characteristics/Synovial fluid/2

Answers to Questions 39–40

- 39. B** Synovial fluid is normally sterile, and all of the organisms listed may cause septic arthritis. *N. gonorrhoeae* is responsible for about 75% of septic arthritis cases occurring in young and middle-aged adults. *Staphylococcus* is responsible for the majority of cases involving older adults and is the most frequently found isolate from infected joint replacements. *Haemophilus* spp., *Staphylococcus* spp., and *Streptococcus*

spp. are the most common causes of arthritis in young children.

- 40. A** The WBC count of normal joint fluid is 200/ μ L or less. Values greater than 5,000/ μ L cause the fluid to be purulent and occur in septic arthritis, RA, and gout. WBC counts greater than 50,000 μ L indicate septic arthritis. The majority of WBCs in normal fluid are monocytes, which usually account for 50% to 65%. Neutrophils and lymphocytes should account for no greater than 25% each. An increase in RBCs occurs in cases of infectious and hemorrhagic arthritis or results from a traumatic tap. Hemorrhagic fluid will appear turbid, red to brown, and often clotted. Inflammatory arthritis can allow fibrinogen to enter the fluid and thus clot. Fluid from a person with hemophilia will not clot in spite of the bloody appearance.

7.4

Amniotic, Gastrointestinal, and Seminal Fluids

1. **Which of the following statements about amniotic fluid bilirubin measured by scanning spectrophotometry is true?**
 - A. The 410-nm peak is caused by hemoglobin and the 450-nm peak is caused by bilirubin
 - B. Baseline correction is not required because a diode array detector is used
 - C. Chloroform extraction is necessary only when meconium is present
 - D. In normal amniotic fluid, bilirubin increases with gestational age

Body fluids/Apply principles of special procedures/Amniotic fluid/2
2. **Which test best correlates with the severity of hemolytic disease of the fetus and newborn (HDFN)?**
 - A. Rh antibody titer of the mother
 - B. Lecithin/sphingomyelin (L/S) ratio
 - C. Amniotic fluid bilirubin
 - D. Urinary estradiol

Body fluids/Correlate clinical and laboratory data/Amniotic fluid/2
3. **Which is the reference method for determining fetal lung maturity (FLM)?**
 - A. Human placental lactogen
 - B. L/S ratio
 - C. Amniotic fluid bilirubin
 - D. Urinary estriol

Body fluids/Correlate laboratory data with physiological processes/L/S ratio/2

Answers to Questions 1–3

1. **A** Amniotic fluid bilirubin reflects the extent of fetal RBC destruction in cases of HDFN. The fluid is centrifuged and absorbance measured from 350 to 600 nm simultaneously by using a photodiode array detector. Background absorbance is corrected by constructing a baseline using the absorbances at 365 nm and 550 nm. The delta absorbance (ΔA) of hemoglobin at 410 nm and of bilirubin at 450 nm is determined by subtracting the absorbance of the baseline from the respective peaks. Samples that are not grossly hemolyzed can be corrected for oxyhemoglobin by subtracting 5% of the ΔA at 410 nm from the ΔA at 450 nm. When hemolysis is severe or meconium is present, bilirubin must be extracted in chloroform before measuring absorbance. Bilirubin normally decreases with increasing gestational age because fetal urine contributes more to amniotic fluid volume as the fetus matures. The bilirubin concentration must be correlated with gestational age to correctly evaluate the severity of HDFN.
2. **C** Amniotic fluid bilirubin is the best index of the severity of HDFN and is measured by diode array spectrophotometry across the 365 to 550 nm range. When hemoglobin

produces a positive slope at 410 nm, the bilirubin should be extracted with chloroform prior to scanning. Extraction methods give the best correlation with RBC destruction.

3. **B** Respiratory distress syndrome (RDS) develops when surfactants are insufficient to prevent collapse of the infant's alveoli during expiration. Tests measuring pulmonary phospholipid surfactants are the most specific and sensitive indicators of RDS. An L/S ratio greater than 2:1 (in some laboratories 2.5:1) is the most widely accepted measure of FLM. Most of the surfactants in the amniotic fluid are present in the form of lamellar bodies. These can be counted using an electronic cell counter at the settings for enumerating platelets. Laboratories must correlate the LBC that corresponds to an LS ratio of 2:1. The LBC cutoff will differ depending on the counter.

4. **Which of the following statements regarding the L/S ratio is true?**

- A. A ratio of 2:1 or greater usually indicates adequate pulmonary surfactant to prevent RDS
- B. A ratio of 1.5:1 indicates fetal lung maturity in pregnancies associated with diabetes mellitus
- C. Sphingomyelin levels increase during the third trimester, causing the L/S ratio to fall slightly during the last 2 weeks of gestation
- D. A phosphatidylglycerol (PG) spot indicates the presence of meconium in the amniotic fluid

Body fluids/Correlate clinical and laboratory data/L/S ratio/2

5. **Which of the following conditions is most likely to cause a falsely low L/S ratio?**

- A. The presence of PG in amniotic fluid
- B. Freezing the specimen for one month at -20°C
- C. Centrifugation at $1,000 \times g$ for 10 minutes
- D. Maternal diabetes mellitus

Body fluids/Apply knowledge to recognize sources of error/Amniotic fluid/3

6. **Which of the following statements accurately describes human chorionic gonadotropin (hCG) levels in pregnancy?**

- A. Levels of hCG rise throughout pregnancy
- B. In ectopic pregnancy, serum hCG doubling time is below expected levels
- C. Molar pregnancies are associated with lower levels than expected for the time of gestation
- D. hCG returns to nonpregnant levels within 2 days following delivery, stillbirth, or abortion

Body fluids/Correlate clinical and laboratory data/Chorionic gonadotropin/2

Answers to Questions 4–6

4. **A** Pulmonary surfactants are mainly disaturated lecithins produced by type II granular pneumocytes. The L/S ratio increases toward the end of the third trimester because of increased production of lecithin. The concentration of sphingomyelin remains constant throughout gestation and serves as an internal reference. Meconium contains less lecithin than amniotic fluid and will usually decrease the L/S ratio; however, meconium produces a spot that can be misinterpreted as lecithin, leading to a falsely increased L/S ratio. Sufficient PG to produce a spot is seen only when the L/S ratio is 2:1 or higher. PG is not present in either blood or meconium, and therefore, its presence indicates fetal lung maturity. In diabetes, the fetal lungs may mature more slowly than normal, and infants may develop RDS when the L/S ratio is 2:1 or slightly

higher. For this reason, an L/S of 3:1 more closely correlates with fetal lung maturity when testing amniotic fluid from mothers with diabetes. As in all other cases, when the amniotic fluid from a diabetic mother is positive for PG, fetal lung maturity is established.

5. **C** Pulmonary surfactants are largely present in the form of lamellar bodies and can be lost by centrifuging the amniotic fluid at high g force. Centrifuge speed should be the minimum required to spin down cells (450 g for 10 minutes at 4°C). Samples that cannot be measured immediately should be refrigerated or frozen. Samples are stable for up to 3 days at 2°C to 8°C and for months when frozen at -20°C or lower. Meconium and blood may also introduce errors when measuring the L/S ratio. Blood has an L/S ratio of approximately 2:1 and will falsely raise the L/S ratio when fetal lungs are immature and depress the L/S ratio when fetal lungs are mature.
6. **B** In normal pregnancy, hCG levels rise exponentially following implantation and peak at weeks 9 to 12, reaching in excess of 100,000 mIU/mL. The hCG doubling time averages 2.2 days. In ectopic pregnancy, the expected increase between consecutive days is below normal. Levels fall after the first trimester to about 20,000 mIU/mL and then remain at about that level through term. Hydatiform moles are associated with greatly elevated levels of hCG. Serum hCG can take up to 4 weeks to return to nonpregnant (less than 25 mIU/mL) or baseline (less than 5 mIU/mL) levels following delivery, stillbirth, or abortion.

7. **Which of the following statements regarding pregnancy testing is true?**

- A. β -Subunits of hCG, thyroid-stimulating hormone (TSH), and follicle-stimulating hormone (FSH) are identical
B. Antibodies against the β -subunit of hCG cross-react with luteinizing hormone (LH)
C. A false-positive result may occur in patients with heterophile antibodies
D. Serum should not be used for pregnancy tests because proteins interfere
- Body fluids/Apply principles of basic laboratory procedures/Pregnancy test/2*

8. **SITUATION: A pregnant woman was seen by her physician, who suspected a molar pregnancy. An hCG test was ordered, and hCG levels were found to be low. The sample was diluted 10-fold and the assay was repeated. The level was found to be grossly elevated. What best explains this situation?**

- A. The wrong specimen was diluted
B. A pipeting error was made in the first analysis
C. Antigen excess caused a falsely low result in the undiluted sample
D. An inhibitor of the antigen-antibody reaction was present in the sample
- Body fluids/Apply knowledge to recognize sources of error/hCG/3*

9. **Which assay result is often approximately 25% below the expected level in pregnancies associated with Down syndrome?**

- A. Serum unconjugated estriol
B. L/S ratio
C. Amniotic fluid bilirubin
D. Urinary chorionic gonadotropin

Body fluids/Correlate laboratory data with physiological processes/Trisomy screening/2

Answers to Questions 7–9

7. **C** The α -subunit of hCG is very similar to the α -subunit of TSH and FSH and identical to LH. Although the β -subunits of hCG and LH are very similar, antibodies can be made to the β -subunit of hCG that do not cross-react with LH or other pituitary hormones. Most enzyme immunoassay (EIA) methods utilize two monoclonal antibodies against different sites of the hCG molecule. One antibody is specific for the carboxy terminal end of the β -chain, and the other reacts with the α -chain, resulting in a positive test only when intact hCG is present. Because monoclonal antibodies are usually derived from mouse hybridomas, rare false-positive results may occur in patients who have anti-mouse immunoglobulin antibodies. Although the test can detect lower levels of hCG, 25 mIU/mL is the positive cutoff for pregnancy. Serum is preferred over urine because serum levels are more consistently above the cutoff compared with random urine in very early pregnancy.
8. **C** Assays of intact hCG are double antibody sandwich immunoassays. One antibody reacts with the α -subunit and the other with the β -subunit. In assays where both antibodies are added together, a process called the “hook effect” is known to occur. In extreme antigen excess, the hCG saturates both antibodies, preventing sandwich formation. This results in measurement of falsely low levels of hCG.
9. **A** Estriol is produced by the placenta as well as the fetal and maternal adrenal glands and liver. Free estriol produced by the placenta is rapidly conjugated by the maternal liver. Almost all of maternal serum unconjugated (free) estriol is derived from the fetus and is a direct reflection of current fetal placental function. Serum unconjugated estriol (uE_3) measured during the second trimester is used along with serum AFP, hCG, and dimeric inhibin A as part of the quad marker screening test for Down syndrome. AFP and uE_3 are decreased by approximately 25%, inhibin A is increased by a factor of approximately 1.8, and hCG is increased by a factor of approximately 2.5 in Down syndrome pregnancies. When all four assays are combined with adjustments for maternal age, gestational age, race, maternal weight, and diabetes, the detection rate is approximately 70% to 80% and the false-positive rate 7%.

10. Which of the following statements about alpha fetoprotein (AFP) is correct?

- A. Maternal serum may be used to screen for open neural tube defects
- B. Levels above 4 ng/mL are considered positive
- C. Elevated levels in amniotic fluid are specific for spina bifida
- D. AFP levels increase in pregnancies associated with Down syndrome

Body fluids/Apply principles of special procedures/Alpha fetoprotein/2

11. First-trimester screening for Down syndrome can be performed using which markers?

- A. AFP and unconjugated estriol
- B. Free β hCG and pregnancy-associated plasma protein A
- C. Intact hCG and dimeric inhibin A
- D. Dimeric inhibin B and AFP

Body fluids/Apply knowledge of special procedures/Trisomy screening/1

12. When performing marker screening tests for Down syndrome, why are results

expressed in multiples of the median (MoM), rather than in concentration?

- A. Concentration is not normally distributed
- B. MoM normalizes for gestational age
- C. Some tests cannot be reported in mass units
- D. Mean cannot be determined accurately for these analytes

Body fluids/Apply knowledge of special procedures/Trisomy screening/1

Answers to Questions 10-12

- 10. A** Maternal serum AFP increases steadily until 2 months prior to full term as AFP passes from the fetal urine into the amniotic fluid. Because serum levels are dependent on gestational age, URLs depend on the date of the last menstrual period. AFP is measured between 14 and 18 weeks gestation, and levels are reported as multiples of the median to permit interlaboratory comparison. When serum levels are high, ultrasonography is used to determine fetal age and rule out twins. Increased maternal serum AFP levels (greater than 2.5 MoM) may result from many diseases in addition to open neural tube defects, such as spina bifida. These include anencephaly, ventral wall defects, congenital hypothyroidism, and Turner syndrome. Decreased levels (less than 0.75 MoM) may be seen in approximately 25% of Down syndrome pregnancies.
- 11. B** Maternal serum AFP levels are too low to measure accurately during the first trimester, and intact hCG and estriol do not discriminate well between 21 trisomy pregnancy and normal pregnancy before the second trimester. First trimester screening for Down syndrome (and trisomy 18) can be performed between weeks 10 and 13 by using free β -hCG (almost twofold higher in Down syndrome) and pregnancy-associated plasma protein A (PAPP-A) which has a median in Down syndrome less than half of that seen in normal pregnancy. These two markers used together with high-resolution ultrasonography to determine nuchal fold thickness (swelling at the base of the neck) have a sensitivity of 85% to 90%. Nuchal fold thickness (NT) in Down syndrome averages 1.5 MoM compared with 1.0 MoM for normal pregnancy.
- 12. B** Reporting of screening markers as MoM has two advantages: (1) It eliminates interlaboratory variation in reference ranges seen when concentration units are reported. Laboratories using different methods (antibodies or calibrators) may have significantly different mass unit results for the same sample, necessitating different reference ranges. (2) The reference range in concentration units is also dependent on the gestational age at the time of sample collection; however, the average result for normals is always 1.0 MoM, regardless of the gestational age of the cohort. Use of MoM obviates the need to report specific reference ranges based on method or gestational age and makes calculation of risk less complicated.

13. Which test for 21 trisomy is *not* recommended for women who are carrying twins?

- A. First trimester triple marker screening
- B. Maternal plasma cell free fetal DNA
- C. Chorionic villus sampling
- D. Second trimester quad marker screening

Body fluids/Apply knowledge to recognize sources of error//Trisomy detection/2

Answer to Question 13

- 13. B** Down syndrome can result from a 14–21 translocation or isochromosome formation, but most cases arise from nondisjunction of chromosome 21 during meiosis. Plasma cell free DNA testing can be done as early as week 10 and detects fetal DNA released from fetal cells that has crossed the placenta. The test measures the ratio of fetal DNA:maternal DNA from chromosomes X, Y, 13, 18, and 21 with use of single nucleotide polymorphisms on the respective chromosomes to distinguish fetal DNA from maternal DNA. This ratio can be misinterpreted if more than one fetus is present. If any screening test is positive, amniocentesis or chronic villus sampling is performed, and trisomy is investigated by using chromosome karyotyping, microarray, or fluorescence in situ hybridization (FISH) on fetal cells.

14. Which statement regarding the fetal fibronectin test is true?

- A. A positive test is correlated with a low probability of delivery within 14 days
- B. The test should not be performed before week 24 or after the end of week 34**
- C. The test is performed on amniotic fluid
- D. The test is used to identify amniotic fluid after rupture of the fetal membranes

Body fluids/Apply principles of special laboratory procedures/Fetal fibronectin/1

15. What is the term for spermatozoa in which the anterior portion of the headpiece is smaller than normal?

- A. Azoospermia
- B. Microcephaly**
- C. Acrosomal deficiency
- D. Necrozoospermia

Body fluids/Apply knowledge of fundamental biological characteristics/Seminal fluid/1

16. What is the most common cause of male infertility?

- A. Mumps
- B. Klinefelter syndrome**
- C. Varicocele
- D. Malignancy

Body fluids/Correlate clinical and laboratory data/Seminal fluid/2

17. Which of the following values is the lower limit of normal for sperm concentration?

- A. 15 million per milliliter
- B. 40 million per milliliter**
- C. 60 million per milliliter
- D. 100 million per milliliter

Body fluids/Evaluate laboratory data to recognize health and disease states/Seminal fluid/2

Answers to Questions 14–17

- 14. B** The fetal fibronectin test is used mainly to rule out the likelihood of preterm delivery in high-risk pregnancies or in women with signs of preterm labor. Fetal fibronectin is a basement membrane protein produced by the amnion and chorion. It is present in cervical secretions in early pregnancy but disappears by about week 20. When there is inflammation to the membranes preceding delivery, fibronectin is released and can be

found in cervicovaginal secretions. A positive test (greater than 50 ng/mL) has a sensitivity of about 60% in predicting preterm birth. However, a negative test has a 92% negative predictive value for the likelihood of preterm delivery and effectively rules out preterm delivery within the next 2 weeks. Amniotic fluid that has escaped from ruptured membranes is identified by testing a vaginal swab for pH. Vaginal fluid is normally acidic, with a pH between 5.0 and 6.0. After rupture of the membranes, the pH of the fluid changes to 6.5 to 7.5. This change can be detected by using Nitrazine paper or a swab containing Nitrazine yellow.

15. **C** Spermatozoa have a well-defined headpiece consisting of the acrosome and nucleus. The acrosome comprises the anterior portion of the head and contains nutrients and enzymes needed for penetration of the ovum. A thin filament, the neckpiece, connects the head and tail. The tail is divided into the midpiece, principal piece (mainpiece), and endpiece. The midpiece is the thick anterior end of approximately 5 μ . It contains a $9 \times 2 + 2$ longitudinal arrangement of microtubules (two central microtubules surrounded by nine doublets so that a cross section resembles a pinwheel). This is called the *axoneme* and it is surrounded radially by nine outer dense fibers which in the midpiece are covered by a sheath containing mitochondria. The longest portion of the tail (40–45 μ) is the principal piece. Its axoneme is thinner than the midpiece. The distal portion, called the *endpiece*, is approximately 5 μ . It contains the axoneme but is unsheathed.
16. **C** Varicocele is the hardening of veins that drain the testes. This causes blood from the adrenal vein to flow into the spermatic vein. Adrenal corticosteroids retard the development of spermatozoa. Mumps, Klinefelter syndrome, and malignancy cause testicular failure, which accounts for about 10% of infertility cases in men.
17. **A** The reference range for spermatozoa is 15 to 150×10^6 /mL. Concentrations below 15×10^6 /mL are considered abnormal. The sperm concentration is multiplied by the seminal fluid volume to determine the sperm count. The lower limit of normal for the sperm count is 40×10^6 per ejaculate. This often results from obstruction of the ejaculatory duct or testicular failure.

18. Which morphological abnormality of sperm is most often associated with varicocele?

- A. Tapering of the head
- B. Cytoplasmic droplet below the neckpiece
- C. Lengthened neckpiece
- D. Acrosomal deficiency

Body fluids/Correlate clinical and laboratory data/Seminal fluid/2

19. Which of the following stains is used to determine sperm viability?

- A. Eosin Y
- B. Hematoxylin
- C. Papanicolaou
- D. Methylene blue

Body fluids/Apply principles of special procedures/Seminal fluid/1

20. Which of the following semen analysis results is abnormal?

- A. Volume 1.0 mL
- B. Liquefaction 40 minutes at room temperature
- C. pH 7.6

D. Motility 50% progressive movement

Body fluids/Evaluate data to recognize abnormal results/Seminal fluid/2

21. Which of the following sample collection and processing conditions will lead to inaccurate seminal fluid analysis results?

A. Sample stored at room temperature for 1 hour before testing

B. Sample collected following coitus

C. Sample collected without an anticoagulant

D. Sample collected without use of a condom

Body fluids/Apply knowledge to recognize sources of error/Seminal fluid/2

Answers to Questions 18–21

- 18. A** Acrosomal deficiency, nuclear abnormalities, and lengthened neckpiece are the most common morphological abnormalities of spermatozoa. Tapering of the head is a nuclear abnormality. Sperm morphology should be evaluated by classifying 200 mature sperm in duplicate by strict criteria. There are several strict criteria in use. The normal sperm head is approximately 4.0 to 5.0 μm in length, 2.5 to 3.5 μm in width, has an L/W ratio of 1.3 to 1.8, and an acrosomal area of 40% to 70%. Using strict criteria, there is a high likelihood of infertility when the number of normal forms is below 4%.
- 19. A** Eosin Y is excluded by living sperm and is used to determine the percentage of living cells. Papanicolaou, Giemsa, and hematoxylin stains are used to evaluate sperm morphology. The viability test should be performed whenever the results of the motility test are subnormal.
- 20. A** The normal volume of seminal fluid is 1.5 to 5.0 mL. A lower volume than 1.5 mL causes a low sperm count ($\text{sperm/mL} \times \text{volume}$) and can be caused by absence of the seminal vesicles or prostate, ductal obstruction, or retrograde ejaculation of seminal fluid into the urinary bladder. The seminal fluid should coagulate within 5 minutes after ejaculation because of secretions of the seminal vesicles. Proteases, such as prostate-specific antigen (PSA) hydrolyze semenogelin and fibronectin, causing liquefaction to occur within 1 hour at room temperature. The seminal fluid pH should be between 7.2 and 8.0. Motility is evaluated by grading the movement of two replicates of 200 sperm in five high-power fields. It is normal when 32% or greater show progressive movement or when 40% or greater show progressive and nonprogressive movement.
- 21. B** A seminal fluid sample should not be collected following coitus. The patient should abstain from ejaculation for at least 2 days but no more than 7 days prior to submitting the sample. A condom should not be used because it may contain spermicides. The sample should be collected at the testing site in a sterile jar with a wide opening, and stored at room temperature. The specimen should be analyzed as soon as possible. The time between collection and delivery to the laboratory must be documented. Motility should be determined as soon as the fluid has liquefied (maximum storage time is 1 hour). Anticoagulants are not used; if the sample fails to liquefy, it can be treated with chymotrypsin before analysis.

22. When performing a seminal fluid analysis, what is the upper limit of normal for WBCs?

- A. $1 \times 10^6/\text{mL}$
- B. $5 \times 10^6/\text{mL}$
- C. $10 \times 10^6/\text{mL}$
- D. $20 \times 10^6/\text{mL}$

Body fluids/Evaluate laboratory data to recognize health and disease states/Seminal fluid/2

23. Which carbohydrate measurement is clinically useful when performing a seminal fluid analysis?

- A. Glucose
- B. Galactose
- C. Fructose
- D. Maltose

Body fluids/Apply knowledge of special procedures/Seminal fluid/1

24. Which condition is most often associated with gastric ulcers?

- A. Cancer of the stomach
- B. *Helicobacter pylori* infection
- C. Zollinger-Ellison (Z-E) syndrome
- D. Pernicious anemia

Body fluids/Gastric/Correlate clinical and laboratory data/2

25. In which condition is the highest level of serum gastrin usually seen?

- A. Atrophic gastritis
- B. Pernicious anemia
- C. Z-E syndrome
- D. Cancer of the stomach

Body fluids/Correlate clinical and laboratory data/Gastric function/2

Answers to Questions 22–25

- 22. A** When evaluating sperm morphology, the number of round cells (immature spermatozoa and WBCs) should also be determined. The number of each is counted along with 200 mature sperm and then divided by 2 to determine their percentage. This is multiplied by the sperm concentration to give the absolute count per milliliter. An increased number of WBCs is an indicator of infection and is usually associated with prostatitis. Round cells are also estimated by noting their number per high-power field. Each round cell per field counted with the $40\times$ objective corresponds to one million per milliliter. The upper limit of normal for WBCs is $1 \times 10^6/\text{mL}$ and for immature sperm $5 \times 10^6/\text{mL}$.
- 23. C** Fructose is the primary nutrient in the seminal fluid and is needed for motility. It is supplied by the seminal vesicles and is low when the vas deferens or seminal vesicles are absent. The lower limit of normal is 150 mg/dL, or 13 μmol , per ejaculate.
- 24. B** Peptic ulcer disease may be caused by either gastric or duodenal ulcers, which are associated with discomfort, hyperacidity, and bleeding. Hyperacidity is most often caused by *H. pylori* infection, which can cause both gastric and duodenal ulcers. In the absence of a positive test for *H. pylori* (e.g., endoscopic biopsy, breath test, enzyme-

linked immunosorbent assay [ELISA], polymerase chain reaction [PCR]) and no history of drug induced ulcers, Z-E syndrome (gastrinoma) should be suspected, and can usually be identified by a plasma gastrin assay. Cancer of the stomach is associated with increased gastric fluid volume but not hyperacidity. Pernicious anemia is associated with gastric hypoacidity, and not ulcers.

25. **C** Gastrin is produced by specialized epithelium of the stomach and stimulates secretion of HCl by parietal cells. Secretion is controlled by negative feedback causing levels to be high in conditions associated with achlorhydria, such as atrophic gastritis. Z-E syndrome results from a gastrin-secreting tumor, gastrinoma, usually originating in the pancreas. It is characterized by very high levels of plasma gastrin and excessive gastric acidity. In duodenal ulcers, increased gastric acidity occurs, but fasting plasma gastrin levels are normal. However, postprandial gastrin levels may be elevated in these patients because they do not respond to the negative feedback signal caused by HCl release. In stomach cancer, gastric volume is increased but acidity is not, and plasma gastrin levels are variable.

26. **In determining free HCl, the gastric fluid is titrated to pH ____.**

- A. 6.5
- B. 4.5
- C. 3.5
- D. 2.0

Body fluids/Apply principles of special procedures/Gastric pH/1

27. **Which test can identify persons with gastrin-secreting tumors that do not demonstrate a definitively increased plasma gastrin concentration?**

- A. Secretin stimulation
- B. Pentagastrin
- C. Cholecystokinin–pancreozymin
- D. Trypsinogen

Body fluids/Select tests/Gastrointestinal function/2

28. **Which of the following tests would be normal in pancreatic insufficiency?**

- A. Secretin stimulation
- B. D-Xylose absorption
- C. Twenty-four-hour fecal fat
- D. β -Carotene absorption

Body fluids/Correlate clinical and laboratory data/Pancreatic function/2

Answers to Questions 26–28

26. **C** Gastric analysis is performed rarely because endoscopic procedures usually are sufficient to diagnose hypo- and hyperacidity states. In difficult cases, analysis of gastric fluid can be used to make a definitive diagnosis. Free HCl in gastric residue from a 12-hour fasting sample obtained by nasogastric suction is measured by titrating with 0.1 N NaOH to a pH 3.5. Total acidity is titrated to pH 7.0, and includes contributions of other acids, including proteins and salts of chloride. Basal acid output (BAO) and peak acid output (PAO) are determined by using timed collection of gastric sample aliquots before and after stimulation of HCl release by pentagastrin. In

achlorhydria, the fasting gastric pH is often greater than 6.0, and this is considered diagnostic. The BAO:PAO ratio is normally less than 0.2. Patients with gastric ulcers may also have a ratio less than 0.2 or between 0.2 and 0.4. In duodenal ulcers, the ratio is usually between 0.2 and 0.6. The ratio is greater than 0.6 only in Z-E syndrome.

- 27. A** Plasma gastrin levels greater than 1,000 pg/mL are usually diagnostic of Z-E syndrome. Smaller elevations can occur in other types of hyperacidity, including gastric ulcers, in renal disease, and after vagotomy. Z-E syndrome can be differentiated from the others by using the secretin stimulation test. Secretin is administered intravenously, and timed plasma samples are collected and measured for gastrin. In Z-E syndrome, at least one specimen should show an increase of 200 pg/mL above the baseline for gastrin. Cholecystokinin–pancreozymin (CCK-PZ), a hormone produced by the small intestine, stimulates HCl production in the stomach, pancreatic release of bicarbonate, and intestinal motility. It may be measured to diagnose intestinal malabsorption, or used along with tests of other gastric regulatory peptides to evaluate dysfunction. Trypsinogen is a precursor of trypsin and is produced by the pancreas. Urinary trypsinogen is increased in acute pancreatitis, whereas fecal trypsin and chymotrypsin are decreased in cystic fibrosis because of pancreatic duct obstruction.
- 28. B** The xylose absorption test differentiates pancreatic insufficiency from malabsorption syndrome (both cause deficient fat absorption). Xylose is absorbed by the small intestine without the aid of pancreatic enzymes. It is not metabolized and is excreted into urine. Low levels indicate gastrointestinal malabsorption.

29. Which of the following is commonly associated with occult blood?

- A.** Colon cancer
- B.** Atrophic gastritis
- C.** Pernicious anemia
- D.** Pancreatitis

Body fluids/Correlate clinical and laboratory data/Occult blood/2

30. Which test is most sensitive in detecting persons with chronic pancreatitis?

- A.** Fecal trypsin
- B.** Fecal chymotrypsin
- C.** Fecal elastin-1
- D.** Plasma lipase

Body fluids/Correlate clinical and laboratory data/Fecal enzymes/2

Answers to Questions 29–30

- 29. A** Blood in feces is a very sensitive indicator of gastrointestinal bleeding and is an excellent screening test to detect asymptomatic ulcers and malignancy of the gastrointestinal tract. However, the test is nonspecific and contamination with vaginal blood is a frequent source of error.
- 30. C** Digestive enzyme tests are used to identify infants and children with pancreatic insufficiency as a consequence of cystic fibrosis, and adults who have chronic pancreatitis. Such tests include fecal trypsin and chymotrypsin, serum immunoreactive trypsin, serum lipase, and fecal elastase-1. Fecal chymotrypsin is somewhat more

sensitive in diagnosing pancreatic insufficiency and chronic pancreatitis than fecal trypsin or serum immunoreactive trypsin, but the most sensitive and specific enzyme test is fecal elastase-1, which is low in about 90% of persons with cystic fibrosis as well as pancreatic insufficiency and in greater than 75% of adults with moderate to severe chronic pancreatitis. Lipase is neither sensitive nor specific for chronic pancreatitis.

7.5

Urinalysis and Body Fluids Problem-Solving

1. **Given the following dry reagent strip urinalysis results, select the *most appropriate course of action*:**

pH = 8.0	Protein = 1+	Glucose = Neg	Blood = Neg
Ketone = Neg		Nitrite = Neg	Bilirubin = Neg

- A. Report the results, assuming acceptable quality control
 B. Check pH with a pH meter before reporting
 C. Perform a turbidimetric protein test, instead of the dipstick protein test, and report
 D. Request a new specimen

Body fluids/Evaluate laboratory data to recognize problems/Urinalysis/3

2. **Given the following urinalysis results, select the *most appropriate course of action*:**

pH = 8.0	Protein = Trace	Glucose = Neg
Ketone = Small	Blood = Neg	Nitrite = Neg

Microscopic findings:

RBCs = 0–2/HPF	WBCs = 20–50/HPF
Bacteria = Large	Crystals = Small, CaCO ₃

- A. Call for a new specimen because urine was contaminated in vitro
 B. Recheck pH because CaCO₃ does not occur at alkaline pH
 C. No indication of error is present; results indicate a UTI
 D. Report all results except bacteria because the nitrite test was negative

Body fluids/Evaluate laboratory data to recognize inconsistent results/Urinalysis/3

3. **SITUATION: A 6-mL pediatric urine sample is processed for routine urinalysis in the usual manner. The sediment is prepared by centrifuging all of the urine remaining after performing the biochemical tests. The following results are obtained:**

SG = 1.015	Blood = Large	Leukocytes = Moderate
Protein = 2+	RBCs: 5–10/HPF	WBCs: 5–10/HPF

Select the *most appropriate course of action*.

- A. Report these results; blood and protein correlate with microscopic results
 B. Report biochemical results only; request a new sample for the microscopic examination
 C. Request a new sample and report as quantity not sufficient (QNS)
 D. Recentrifuge the supernatant and repeat the microscopic examination

Body fluids/Apply knowledge to recognize sources of error/Urinalysis/3

1. **C** Highly buffered alkaline urine may cause a false-positive result on the dry reagent strip protein test by titrating the acid buffer on the reagent pad. Protein should be measured by an alternate method that is not subject to positive interference by highly buffered alkaline urine.
2. **C** A positive nitrite result requires infection with a nitrate-reducing organism, dietary nitrate, and incubation of urine in the bladder. The test is positive in about 70% of UTI cases. Alkaline pH, bacteriuria, and leukocytes point to UTI.
3. **B** This discrepancy between the blood reaction and RBC count resulted from spinning less than 12 mL of urine. When volume is below 12 mL, the sample should be diluted with saline to 12 mL before concentrating. Results are multiplied by the dilution (12 mL/mL urine) to give the correct range.

4. Given the following urinalysis results, select the *most appropriate* course of action:

pH = 6.5	Protein = Neg	Glucose = Neg
Ketone = Trace	Blood = Neg	Bilirubin = Neg

Microscopic findings:

Mucus = Small	Ammonium urate = Large
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- A. Recheck urine pH
- B. Report these results, assuming acceptable quality control
- C. Repeat the dry reagent strip tests to confirm the ketone result
- D. Request a new sample and repeat the urinalysis

Body fluids/Evaluate laboratory data to recognize problems/Urinalysis/3

5. Given the following urinalysis results, select the *most appropriate first* course of action:

pH = 6.0	Protein = Neg	Glucose = Neg
Ketone = Neg	Blood = Neg	Bilirubin = Neg

Other findings:

Color: Intense yellow	Transparency: Clear	Microscopic: Crystals, Bilirubin granules = Small
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- A. Repeat the dry reagent strip test for bilirubin
- B. Request a new sample
- C. Recheck the pH
- D. Perform a test for urinary urobilinogen

Body fluids/Evaluate laboratory data to determine possible inconsistent results/Urinalysis/3

6. A biochemical profile gives the following results:

Creatinine = 1.4 mg/dL
BUN = 35 mg/dL
K = 5.5 mmol/L

All other results are normal, and all tests are in control. Urine from the patient has an osmolality of 975 mOsm/kg. Select the *most appropriate* course of action.

- A. Check for hemolysis

- B. Repeat the BUN, and report only if normal
- C. Repeat the serum creatinine, and report only if elevated
- D. Report these results

Body fluids/Evaluate laboratory data to recognize problems/Renal function/3

7. **A 2 p.m. urinalysis shows trace glucose on the dry reagent strip test. Fasting blood glucose drawn 8 hours earlier is 100 mg/dL. No other results are abnormal. Select the most appropriate course of action.**
- A. Repeat the urine glucose, and report if positive
 - B. Perform a test for reducing sugars, and report the result
 - C. Perform a quantitative urine glucose; report as trace if greater than 100 mg/dL
 - D. Request a new urine specimen

Body fluids/Evaluate laboratory data to determine possible inconsistent results/Glucose/3

Answers to Questions 4–7

- 4. **A** Ammonium urate crystals occur at alkaline pH only. The pH should be checked, and if it is below 7.0, the crystals should be reviewed for correct identification. The trace ketone does not require confirmation, provided that the quality control of the reagent strips is acceptable.
 - 5. **A** Bilirubin crystals cannot occur in urine without bilirubin. The dry reagent strip test should be repeated, and if the result is negative, a new specimen should be requested as bilirubin may have degraded due to storage. If a second sample yields a negative result, the crystals should be reviewed before reporting. Abnormal crystals occur only in acid or neutral urine.
 - 6. **D** Patients with prerenal failure usually have a BUN:creatinine ratio greater than 20:1. Reduced renal blood flow causes increased urea reabsorption and high urine osmolality. Patients are usually hypertensive and show fluid retention and hyperkalemia.
 - 7. **A** Urine glucose concentration is dependent on blood glucose concentration at the time urine is formed. The postprandial glucose (2 p.m.) level exceeded the renal threshold, resulting in trace glycosuria. Tests for reducing sugars are not used to confirm a positive urine glucose test result.
8. **Following a transfusion reaction, urine from a patient gives positive test results for blood and protein. The SG is 1.015. No RBCs or WBCs are seen in the microscopic examination. These results:**
- A. Indicate renal injury induced by transfusion reaction
 - B. Support the finding of an extravascular transfusion reaction
 - C. Support the finding of an intravascular transfusion reaction
 - D. Rule out a transfusion reaction caused by RBC incompatibility
- Body fluids/Correlate clinical and laboratory data/Urinalysis/3*
9. **A urine sample taken after a suspected transfusion reaction has a positive test result for blood, but intact RBCs are not seen on microscopic examination. Which test result would rule out an intravascular hemolytic transfusion reaction?**
- A. Negative urine urobilinogen

- B. Serum unconjugated bilirubin below 1.0 mg/dL
- C. Serum potassium below 6.0 mmol/L
- D. Normal plasma haptoglobin

Body fluids/Select routine laboratory procedures to verify test results/Transfusion reaction/3

10. Given the following urinalysis results, select the *most appropriate* course of action:

pH = 5.0	Protein = Neg	Glucose = 1,000 mg/dL
Blood = Neg		Bilirubin = Neg
Ketone = Moderate		SSA protein = 1+

- A. Report the SSA protein test result instead of the dry reagent strip test result
- B. Call for a list of medications administered to the patient
- C. Perform a quantitative urinary albumin
- D. Perform a test for microalbuminuria

Body fluids/Evaluate laboratory data to determine possible inconsistent results/Urinalysis/3

11. Urinalysis results from a 35-year-old woman are as follows:

SG = 1.015	pH = 7.5	Protein = Trace
Glucose = Small	Ketone = Neg	Blood = Neg
Leukocytes = Moderate		

Microscopic findings:

RBCs: 5–10/HPF WBCs: 25–50/HPF

Select the *most appropriate* course of action.

- A. Recheck the blood reaction; if negative, look for budding yeast
- B. Repeat the WBC count
- C. Report all results except that for blood
- D. Request the list of medications used

Body fluids/Evaluate laboratory data to recognize sources of error/Urinalysis/3

Answers to Questions 8–11

- 8. **C** RBCs usually remain intact at a SG of 1.015. The absence of RBCs, WBCs, and casts points to hemoglobinuria caused by intravascular hemolysis rather than glomerular injury. A positive protein reaction will occur if sufficient hemoglobin is present.
- 9. **D** The plasma free hemoglobin will be increased immediately after a hemolytic transfusion reaction, and the haptoglobin will be decreased. The hemoglobin will be eliminated by the kidneys, but the haptoglobin will remain low or undetectable for 2 to 3 days. Normal urine urobilinogen and serum unconjugated bilirubin help in ruling out extravascular hemolysis. Pretransfusion potassium is needed to evaluate the contribution of hemolysis to the post-transfusion result.
- 10. **B** The combination of glucose and ketone positivity of urine points to a patient with insulin-dependent diabetes. A false-positive SSA test result is likely if tolbutamide (Orinase) has been administered.
- 11. **A** A nonhemolyzed trace may have been overlooked and the blood test should be

repeated. A false-negative result (e.g., megadoses of vitamin C) rarely occurs. Yeast cells often accompany pyuria and glycosuria and are easily mistaken for RBCs.

12. A routine urinalysis gives the following results:

pH =6.5	Protein = Neg	Blood = Neg
Glucose= Trace	Ketone = Neg	
Microscopic findings:		
Blood casts:	Mucus:	Crystals: Large,
5–10/LFP	Small	amorphous

These results are most likely explained by:

- A. False-negative blood reaction
- B. False-negative protein reaction
- C. Pseudocasts of urate mistaken for true casts
- D. Mucus mistaken for casts

Body fluids/Evaluate laboratory data to determine possible inconsistent results/Urinalysis/3

13. SITUATION: When examining a urinary sediment under 400× magnification, the medical laboratory scientist (MLS) noted many RBCs to have cytoplasmic blebs and an irregular distribution of the hemoglobin. This phenomenon is most often caused by:

- A. Intravascular hemolytic anemia
- B. Glomerular disease
- C. Hypotonic or alkaline urine
- D. Severe dehydration

Body fluids/Correlate clinical and laboratory data/Hematuria/2

14. SITUATION: A urine specimen is dark orange and turns brown after storage in the refrigerator overnight. The MLS requests a new specimen. The second specimen is bright orange and is tested immediately. Which test result would differ between the two specimens?

- A. Ketone
- B. Leukocyte esterase
- C. Urobilinogen
- D. Nitrite

Body fluids/Apply knowledge to recognize sources of error/Urine color/3

15. A patient's random urine sample consistently contains a trace of protein but no casts, cells, or other biochemical abnormality. The first voided morning sample is consistently negative for protein. These findings can be explained by:

- A. Normal diurnal variation in protein loss
- B. Early glomerulonephritis
- C. Orthostatic or postural albuminuria
- D. Microalbuminuria

Body fluids/Evaluate laboratory data to determine possible inconsistent results/Urinalysis/3

12. C At pH 6.5, amorphous crystals are most often urate. These form yellow- or reddish-brown refractile deposits sometimes resembling blood or granular casts. The number of blood casts reported could not have occurred with negative protein and blood test results.
13. B When RBCs pass through the damaged endothelial wall of the glomerulus, they become distorted, and such cells are described as dysmorphic in appearance. They are characterized by uneven distribution of hemoglobin, cytoplasmic blebs, and an asymmetrical membrane distinct from crenation. The cytoplasm may be extruded from the cell and may aggregate at the membrane giving the cell a wavy appearance. A predominance of dysmorphic RBCs in the microscopic examination points to glomerular bleeding as opposed to hematuria from other causes. Intravascular hemolytic anemia causes hemoglobinuria, rather than hematuria. RBCs lyse in hypotonic and alkaline urine. Severe dehydration is not a cause of hematuria.
14. C Urinary urobilinogen is increased in persons with extravascular hemolysis or hepatocellular liver disease. A freshly voided specimen is needed to detect urobilinogen because it is rapidly photo-oxidized to urobilin. This is accompanied by a color change from orange to brown. Urobilin does not react with 2,4 dimethylaminobenzaldehyde or 4-methoxybenzene diazonium tetrafluoroborate, which are used to detect urobilinogen. Consequently, the urobilinogen test in the first sample will be normal but will be increased in the second sample if tested immediately after collection. The best sample for detecting urobilinogen is a 2-hour timed urine sample collected in the midafternoon, when urobilinogen excretion is highest. Ketones and nitrites do not alter the pigment of the urine sample. Leukocytes cause the urine to be turbid but do not cause abnormal color. These three tests are stable for 24 hours when urine is refrigerated within 30 minutes of collection.
15. C Protein and other constituents of urine will often be highest in the first morning void. A normal first-voided sample makes glomerular disease highly unlikely. Orthostatic albuminuria is a benign condition sometimes seen in adolescents who are tall and have a bent posture that puts back pressure on the kidneys. The quantity of albumin excreted into the urine is small. Diagnosis is made by demonstrating a positive test after the person is erect for several hours, and the absence of proteinuria when the person is recumbent. Microalbuminuria seen in persons with diabetes is usually accompanied by a positive test result for urinary glucose.
16. A urine sample with a pH of 8.0 and a specific gravity of 1.005 had a small positive blood reaction but is negative for protein, and no RBCs are present in the microscopic examination of urinary sediment. What *best* explains these findings?
- A. High pH and low SG caused a false-positive blood reaction
 - B. The blood reaction and protein reaction are discrepant
 - C. Hemoglobin is present without intact RBCs because of hemolysis
 - D. An error was made in the microscopic examination
- Body fluids/Evaluate laboratory data to determine possible inconsistent results/Urinalysis/3*
17. A urine sample has a negative blood reaction and 5 to 10 cells per high-power field that resemble RBCs. What is the *best* course of action?
- A. Mix a drop of sediment with 1 drop of WBC counting fluid and re-examine

- B. Report the results without further testing
 - C. Repeat the blood test, and if negative, report the results
 - D. If the leukocyte esterase test is positive, report the cells as WBCs
- Body fluids/Apply knowledge to recognize sources of error/Microscopic/3*

18. A toluidine blue chamber count on CSF gives the following values:

CSF Counts	Peripheral Blood Counts
WBCs $10 \times 10^6/L$	WBCs $5 \times 10^9/L$
RBCs $1,000 \times 10^6/L$	RBCs $5 \times 10^{12}/L$

After correcting the WBC count in CSF, the MLS should next:

- A. Report the WBC count as $9 \times 10^6/L$ without additional testing
- B. Report the WBC count and number of PMNs identified by the chamber count
- C. Perform a differential on a direct smear of the CSF
- D. Concentrate CSF using a cytocentrifuge and perform a differential

Body fluids/Apply knowledge of standard operating procedures/Cerebrospinal fluid/3

19. A blood-tainted pleural fluid is submitted for culture. Which test result would be most conclusive in classifying the fluid as an exudate?

	Test	Result
A.	LD fluid/serum	0.65
B.	Total protein	3.2 g/dL
C.	RBC count	10,000/ μL
D.	WBC count	1500/ μL

Body fluids/Correlate clinical and laboratory data/Pleural fluid/3

Answers to Questions 16–19

- 16. C** RBCs will lyse in alkaline or dilute urine, and this sample displays both. The blood test is sensitive to as little as 0.015 mg/dL hemoglobin, and the protein test is significantly less sensitive. As a result, trace to small positive blood and negative protein are commonly encountered.
- 17. A** When 5 to 10 RBCs/HPFs are seen, the blood test should show a nonhemolyzed trace. Therefore, it is likely that the cells are not RBCs. RBCs are easily confused with nonbudding yeast and may also be mistaken for WBCs. RBCs will lyse in dilute acetic acid, but yeast and WBCs will not. If a yeast infection is present, then the leukocyte esterase test result will likely be positive; therefore, the leukocyte esterase test cannot be used to determine the identity of the cells. The Sternheimer-Malbin stain can be used to differentiate WBCs from RBCs and yeast.
- 18. D** A differential is performed by using CSF concentrate on all neonatal samples and whenever the WBC count is greater than 5/ μL . A toluidine blue chamber count of PMNs is not sufficiently sensitive to detect neutrophilic pleocytosis.

19. A A traumatic tap makes classification of fluids difficult on the basis of cell counts and protein. The values reported for protein, RBCs, and WBCs can occur in either an exudate or bloody transudate, but the LD ratio is significant.

20. A pleural fluid submitted to the laboratory is milky in appearance. Which test would be most useful in differentiating between a chylous and pseudochylous effusion?

- A. Fluid to serum triglyceride ratio
- B. Fluid WBC count
- C. Fluid total protein
- D. Fluid:serum LD ratio

Body fluids/Select test/Pleural fluid/3

21. A CSF sample from an 8-year-old child with a fever of unknown origin was tested for glucose, total protein, lactate, and IgG index. Glucose was 180 mg/dL, but all other results were within the reference range. The CSF WBC count was $9 \times 10^6/L$, and the RBC count was $10 \times 10^6/L$. The differential showed 50% lymphocytes, 35% monocytes, 10% macrophages, 3% neutrophils, and 2% neuroectodermal cells. What is the most likely cause of these results?

- A. Aseptic meningitis
- B. Traumatic tap
- C. Subarachnoid hemorrhage
- D. Hyperglycemia

Body fluids/Apply knowledge to recognize inconsistent results/CSF/3

22. A WBC count and differential performed on ascites fluid gave a WBC count of 20,000/ μL with 90% macrophages. The gross appearance of the fluid was described by the MLS as “thick and bloody.” It was noted on the report that several clusters of these cells were observed and that the majority of the cells contained many vacuoles resembling paper-punch holes. What do the observations above suggest?

- A. Malignant mesothelial cells were counted as macrophages
- B. Adenocarcinoma from a metastatic site
- C. Lymphoma infiltrating the peritoneal cavity
- D. Nodular sclerosing type Hodgkin disease

Body fluids/Apply knowledge to recognize inconsistent results/Serous fluids/3

23. Given the following data for creatinine clearance, select the most appropriate course of action.

Volume = 2.8 L/day; surface area = $1.73 m^2$; urine creatinine = 100 mg/dL; serum creatinine = 1.2 mg/dL

- A. Report a creatinine clearance of 162 mL/min
- B. Repeat the urine creatinine; results point to a dilution error
- C. Request a new 24-hour urine sample
- D. Request the patient’s age and gender

Body fluids/Evaluate laboratory data to recognize problems/Creatinine clearance/3

Answers to Questions 20–23

20. A Chylous effusions are caused by extravasation of lymphatic fluid into the pleural cavity. Pseudochylous effusions are caused by necrosis. Both fluids often appear white

and opalescent, but both effusions can also be bloody, green, or yellow, in addition to being turbid. However, chylous effusions are odorless and have a twofold higher triglyceride compared with plasma. They also usually show a lymphocytosis. Pseudochylous effusions are foul smelling and usually have a mixed cellularity and an elevated cholesterol. These effusions may have triglyceride, but it is usually below 50 mg/dL. Chylous effusions are most often caused by lymphoma or other malignancy or trauma, and, like pseudochylous effusions, may have an increased LD fluid:serum ratio, total protein, and WBC count.

21. **D** CSF glucose is approximately 60% of the plasma glucose but may be somewhat lower in a person with diabetes. The reference range is approximately 40 to 70 mg/dL. A CSF glucose level above 70 mg/dL is caused by a high plasma glucose that equilibrated with CSF. Therefore, hyperglycorrhachia is caused by hyperglycemia. The WBC count in a child between 5 and 12 years of age is 0 to $10 \times 10^6/L$ (0–10/ μL). The normal RBC count and protein rule out subarachnoid hemorrhage and traumatic tap. Although aseptic meningitis cannot be ruled out conclusively, it is unlikely given a normal WBC count and IgG index.
22. **A** Bloody, exudative fluids with a preponderance of a singular cell type are suggestive of malignancy. The cellularity in malignancy is variable but lymphocytosis occurs in about half the cases. Mesothelial cells normally comprise less than 10% of the cells in serous fluid. They may be resting cells, reactive, degenerated, or phagocytic in nonmalignant conditions. In inflammatory conditions, they are often increased and resemble macrophages. However, clusters or balls of such cells and paper-punch vacuoles throughout the cytoplasm and over the nucleus are characteristics of malignant mesothelial cells. Such cells secrete hyaluronic acid, making the fluid highly viscous. The gross appearance of this fluid suggests malignancy. The description of these cells points to mesothelioma, and this specimen should be referred for cytological examination to confirm the diagnosis.
23. **C** A calculated clearance in excess of 140 mL/min is greater than the upper physiological limit. The high volume per day suggests addition of H_2O , or urine that should have been voided and discarded at the start of sample collection. The result should be considered invalid.

24. **An elevated amylase is obtained on a stat serum collected at 8 p.m. An amylase performed at 8 a.m. that morning was within normal limits. The MLS also noted that urine amylase was measured at 6 p.m. Select the *most* appropriate course of action.**

- A. Repeat the stat amylase; report only if within normal limits
 - B. Repeat both the morning and afternoon serum amylase, and report only if they agree
 - C. Request a new specimen; do not report results of the stat sample
 - D. Review the amylase result on the 6 p.m. urine sample; if elevated, report the stat amylase
- Body fluids/Apply knowledge to recognize inconsistent results/Amylase/3*

25. **Results of an FLM study from a patient with diabetes mellitus are as follows:**

L/S ratio = 2.0; Phosphatidyl glycerol = Neg; Creatinine = 2.5 mg/dL

Given these results, the MLS should:

- A. Report the result and recommend repeating the L/S ratio in 24 hours

- B. Perform scanning spectrophotometry on the fluid to determine if blood is present
- C. Repeat the L/S ratio after 4 hours and report those results
- D. Report results as invalid

Body fluids/Correlate laboratory data to verify test results/L/S ratio/3

26. A 24-hour urine sample from an adult submitted for catecholamines gives a result of 140 µg/day (upper reference limit 150 µg/day). The 24-hour urine creatinine level is 0.6 g/day. Select the best course of action.

- A. Check the urine pH to verify that it is less than 2.0
- B. Report the result in µg catecholamines per milligram of creatinine
- C. Request a new 24-hour urine sample
- D. Measure the VMA, and report the catecholamine result only if elevated

Body fluids/Evaluate laboratory data to recognize problems/Catecholamines/3

27. A sperm motility test was performed and 200 sperm were evaluated in each of two duplicates. The first sample showed progressive movement in 50% and nonprogressive movement in 35%, and 15% were immotile. The second showed progressive movement in 35% and nonprogressive movement in 35%, and 30% were immotile. What is the best course of action?

- A. Report the average of the two values for progressive movement
- B. Report the higher of the two values
- C. Repeat the motility test
- D. Call for a new specimen

Body fluids/Evaluate laboratory data to recognize problems/Seminal fluid/3

Answers to Questions 24–27

- 24. D** Serum amylase peaks 2 to 10 hours after an episode of acute pancreatitis, and this may have caused the elevated serum amylase at 8 p.m. Urinary amylase parallels serum amylase; therefore, a positive urine test at 6 p.m. makes sample collection error unlikely.
- 25. A** In patients with poorly controlled diabetes, lung maturity may be delayed and an L/S ratio of 2:1 may be associated with respiratory distress syndrome. A positive PG spot correlates with an L/S ratio of 2:1 or higher and rules out a falsely increased result caused by blood contamination. However, the appearance of PG in amniotic fluid is often delayed in diabetes. The best course of action is to wait an additional 24 hours and perform another L/S ratio on a fresh sample of amniotic fluid because an L/S ratio of 3:1 would indicate a high probability of lung maturity.
- 26. C** Urine creatinine of less than 0.8 g/day indicates incomplete sample collection. The patient's daily catecholamine excretion would be misinterpreted from this result.
- 27. C** Sperm motility should be performed in duplicate and 200 cells per sample in several fields should be evaluated for movement. Agreement between duplicates must be within a specified percentage based on the category with the highest percentage. In this case, progressive movement is highest at 50%, and the difference between replicates should be no greater than 10%.

28. A quantitative serum hCG is ordered on a male patient. The technologist should:

- A. Perform the test and report the result
- B. Request that the order be cancelled
- C. Perform the test and report the result if negative
- D. Perform the test and report the result only if greater than 25 IU/L

Body fluids/Apply knowledge of standard operating procedures/Human chorionic gonadotropin/3

29. SITUATION: A lamellar body count (LBC) was performed on an amniotic fluid sample that was slightly pink within 1 hour of specimen collection. The sample was stored at 4°C prior to analysis. The result was 25,000/μL, classified as intermediate risk of RDS. The physician waited 24 hours and collected a new sample that was counted within 2 hours of collection on the same instrument. The LCB count of the new sample was 14,000/μL and the patient was reclassified as high risk for delivery. Which statement best explains these results?

- A. Loss of lamellar bodies occurred in the second sample because of storage
- B. Blood caused a falsely elevated result for the first sample
- C. The fetal status changed in 24 hours because of respiratory illness
- D. The difference in counts is the result of day-to-day physiological and instrument variance

Body fluids/Apply knowledge to identify sources of error/Specimen collecting and handling/3

30. When testing for drugs of abuse in urine, which of the following test results indicate dilution and would be cause for rejecting the sample?

- A. Temperature at sample submission 92°F
- B. SG 1.002; creatinine 15 mg/dL,
- C. pH 5.8; temperature 94°C
- D. SG 1.012; creatinine 25 mg/dL

Body fluids/Apply knowledge to identify sources of error/Specimen collecting and handling/3

Answers to Questions 28–30

- 28. A** hCG may be produced in men by tumors of trophoblastic origin, such as teratoma and seminoma, and is an important marker for nontrophoblastic tumors, as well.
- 29. B** Lamellar bodies are small particles containing pulmonary surfactants that are made by type II pneumocytes, and their number in amniotic fluid increases as the concentration of phospholipids increases. They are about the same size as platelets and are counted in the platelet channel of cell counters. If the amniotic fluid sample is contaminated with blood, platelets will falsely raise the LBC. Amniotic fluid samples for LBC are stable for several days when stored at 4°C. However, cutoffs for FLM need to be established by each laboratory because there are significant differences in LBCs between different counters.
- 30. B** Tampering with a sample submitted for abuse substance testing can be either by dilution or substitution. Substance Abuse and Mental Health Services Administration (SAMHSA) certified workplace drug testing laboratories are required to test for both, and reject samples based on SAMHSA cutoffs. A specimen is too dilute for testing if the creatinine is less than 20 mg/dL and SG below 1.003. A sample is considered substituted if creatinine is less than 5.0 mg/dL and SG less than 1.002. Values in A and C for pH and temperature are within acceptable limits.

- 31. SITUATION: A urine specimen has an SG of 1.025 and is strongly positive for nitrite. All other dry reagent strip test results are normal, and the microscopic examination was unremarkable, showing no WBCs or bacteria. The urine sample was submitted as part of a pre-employment physical examination that also includes drug testing. Which most likely caused these results?**
- A. A viral infection of the kidney
 - B. A urinary tract infection in an immunosuppressed person
 - C. An adulterated urine specimen
 - D. Error in reading the nitrite pad caused by poor reflectometer calibration
- Body fluids/Apply knowledge to identify sources of error/Drug testing/3*
- 32. A CSF sample submitted for cell counts has a visible clot. What is the best course of action?**
- A. Count RBCs and WBCs manually after diluting the fluid with normal saline
 - B. Tease the cells out of the clot before counting, then dilute with WBC counting fluid
 - C. Request a new sample
 - D. Perform a WBC count without correction
- Body fluids/Apply knowledge to identify the best course of action/CSF/3*
- 33. Total hemolytic complement and glucose are ordered on a synovial fluid sample that is too viscous to pipet. What is the best course of action?**
- A. Dilute the sample in saline
 - B. Add 1 mg/mL hyaluronidase to the sample, and incubate at room temperature for 30 minutes
 - C. Warm the sample to 65°C for 10 minutes
 - D. Request a new specimen
- Body fluids/Apply knowledge to identify the best course of action/Synovial fluid/3*
- 34. A CSF cytospin smear shows many smudge cells and macrophages with torn cell membranes. What most likely caused this problem?**
- A. Failure to add albumin to the cytospin cup
 - B. Failure to collect the CSF in EDTA
 - C. Centrifuge speed too low
 - D. Improper alignment
- Body fluids/Apply knowledge to identify sources of error/Fluid cell counts/3*

Answers to Questions 31–34

- 31. C** Urine validity testing for drugs of abuse includes tests for nitrite, glutathione, pyridinium dichromate, and peroxide in addition to pH, SG, and creatinine. These substances are known to cause negative interference in the enzyme-multiplied immunoassay technique (EMIT test). A viral infection of the kidney would be associated with high numbers of renal tubular epithelial cells and leukocytes. An infection in an immunosuppressed person would still produce urinary WBCs. Although laboratory error is possible, a false-positive result caused by reflectometer error would be suspected if the test pad were negative when reading it manually.
- 32. C** Although the sample is still suitable for microbiological analysis, it will not give reliable cell counts or biochemistry results because of extensive contamination with

peripheral blood.

- 33. B** Joint fluid too viscous to pipet accurately cannot be diluted accurately. Complement is heat labile, and total hemolytic complement is destroyed when the sample is heated to 56°C for 5 minutes. Joint fluid is difficult to collect, and a new sample is likely to have the same problem.
- 34. A** Cellular distortion caused by centrifugation is a common problem and can be reduced by adding 22% bovine albumin or 10% dextran to the cytospin cup along with the sample. In the absence of a large subarachnoid hemorrhage, CSF does not clot because it contains no fibrinogen, and the sample can be collected and counted without anticoagulant.

35. An automated electronic blood cell counter was used to count RBCs and WBCs in a turbid pleural fluid sample. The WBC count was $5 \times 10^{10}/L$ (50,000/ μL) and the RBC count was $5.5 \times 10^{10}/L$ (55,000/ μL). What is the significance of the RBC count?

- A.** The RBC count is not significant and should be reported as 5,000/ μL
B. The RBC count should be reported as determined by the analyzer
C. A manual RBC count should be performed
D. A manual RBC and WBC count should be performed and reported instead

Body fluids/Apply knowledge to identify sources of error/Fluid cell counts/3

Answer to Question 35

- 35. A** Electronic cell counters are validated for body fluid cell counts with specific minimum detection limits. For most counters, this is 50 to 200 total nucleated cells (TNCs)/ μL and 10,000 RBC/ μL . Specialized counters using fluorescent and image analysis can achieve low end sensitivities for both TNC and RBC counts of 0 to 1/ μL . Because the RBCs are lysed in the WBC bath, the WBC count represents the number of nucleated cells present. When cell counters that perform CBCs are used, the WBCs are not lysed in the RBC bath and would be counted as RBCs. In this case, the empyemic fluid would cause the RBC count to be erroneously elevated, and this should be corrected before reporting by subtracting the WBC count from the RBC count.

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